

Grammaticality de-idealized

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Abstract

This paper introduces the Operator-Value Model of Grammaticality (OVMG), a framework for (un)grammaticality distinct from grammars focused solely on language production or description. The model treats (un)grammaticality as conditioned stability in form–value relations accepted or rejected within communicative situations, integrating mapping viability, operator-value coherence, population licensing, and speaker-level feelings of anomaly. Previous approaches, including generative grammar, Construction Grammar, and psycholinguistic models, have each captured part of the picture but haven’t jointly explained why semantically coherent constructions may nonetheless be deemed ungrammatical, why certain transparent structures remain systematically unentrenched, and why categorical grammaticality clusters where it does. The framework derives categoricity from conventionalized norms, opportunity structure, and preemption dynamics within communicative situations, rather than from structural membership or intuition alone. It earns its keep by projection: it predicts which rejected constructions should soften under repeated exposure and which should resist, how learners should treat zero evidence under different opportunity structures, how early second-language rejections should track first-language preemption, and where community treatment of a form should change discontinuously as licensing crosses a situational threshold.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization; usage-based grammar

INTRODUCTION

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

- (1) a. **Can the have running?*
- b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
- c. **I've finished it yesterday.*
- d. *?I saw Joan, a friend of whose was visiting.*
- e. *The bread the baker the apprentice helped made is delicious.*
- f. A: *How old are you?* B: **I have 25 years.*
- g. **Which did you buy car?*

While all might receive asterisks in many analyses, they represent fundamentally different types of unacceptability. Some constructions, like (1a), fail to establish any form–value relationship in English. Others, like (1c), involve a clash between the time semantics of clause tense and the word *yesterday*. Still others, such as (1d), show gradient or indeterminate status. (1e) illustrates what might be called theoretically predicted acceptability: constructions that linguistic theory predicts should be grammatical but that are consistently rejected by native speakers in ordinary judgment tasks. Still others, like (1f), are transparent and grammatical in related languages but lack licensing in English age predication. And a few, like the left-branch extraction of (1g), seem to be ruled out entirely, despite being apparently short and interpretable.

Different theoretical traditions have emphasized different aspects of grammaticality: formal approaches focus on abstract principles, usage-based theories stress frequency and entrenchment, and processing accounts highlight cognitive constraints. But the field still lacks a unified framework that can explain both categorical constraints and gradient acceptability while accounting for cross-linguistic variation and change over time.

The de-idealization in the title is specific. Early generative grammar idealized grammaticality to the knowledge of an ideal speaker–listener in a completely homogeneous speech community (Chomsky, 1965, p. 3), treating variation, uncertainty, and change as noise around a categorical competence fact. That idealization bought tractability, but it priced out the phenomena in (1). Removing it means asking first what the category GRAMMATICAL lets us predict – its projectibility, what observing some features licenses us to expect about others (Goodman, 1955) – and only then asking what worldly profile supports those predictions and what stabilizes that profile (Boyd, 1999; Reynolds, 2026a).

This paper presents such a framework, building on insights from both generative and functional traditions to propose that grammaticality should be understood in terms of form–value relations that evolve within specific communicative situations. These relations interact with both universal processing constraints and sociolinguistic factors to produce the complex patterns of acceptability observed in actual language use. The framework rests on three key premises:

1. At its core, grammaticality involves conventional form–value relations within specific communicative situations (encompassing communities, dialects, and registers).

2. These relations interact systematically with processing constraints, sociolinguistic factors, and other linguistic subsystems (phonology, semantics, pragmatics).
3. Grammaticality reflects the degree to which an utterance establishes stable, conventionalized form–value relations within the relevant communicative situation. Various factors can disrupt this stability – from complete absence of a licensable analysis to conflicts between competing construals to lack of situational licensing – producing a spectrum of ungrammaticality effects unified by form–value instability.

These premises directly address key limitations in previous approaches to grammaticality. The first premise moves beyond the abstract competence model of early generative grammar while building on its analyses of systematic constraints. It provides a theoretical foundation for explaining both categorical constraints like (1g) and gradient acceptability like (1d), aligning with Francis’s (2022) arguments for incorporating gradience into core grammatical theory. The second premise resolves a long-standing tension between formal syntactic approaches and processing-based accounts by explicitly modeling their interaction rather than treating them as competing explanations. This explains why some ungrammatical constructions resist improvement through exposure while others become more acceptable with familiarity. The third premise provides a principled basis for phenomena that have resisted unified explanation in both generative and functional frameworks.

Section 1 examines the impasse in grammaticality theory, tracing how different theoretical traditions – from early generative grammar through contemporary experimental approaches – have attempted to reconcile form, meaning, and acceptability. Section 2 presents the framework in detail, defining the three constitutive variables that determine grammaticality and identifying recurrent diagnostic instability modes that govern judgments. Section 3 presents a simple formal model that illustrates how the framework’s key components interact to produce grammaticality judgments. Section 5 explores the theoretical implications of this approach, particularly its relationship to generative grammar, Construction Grammar, and usage-based theories, and includes specific predictions about cross-linguistic variation in grammaticality judgments and the conditions under which satiation effects should occur. The paper concludes by acknowledging limitations of the current framework and suggesting directions for future research using corpus analysis, experimental methods, and cross-linguistic investigation.

1 THE IMPASSE IN GRAMMATICALITY THEORY

The concept of grammaticality remains elusive despite its centrality to linguistic theory. After decades of research, the field still lacks a comprehensive account of what makes an utterance grammatical or ungrammatical. This theoretical impasse stems from trying to treat grammaticality as a unified phenomenon, when it actually arises from the interaction of three distinct sources of failure. To understand why previous approaches have struggled, it is necessary to disentangle three components: structural viability, interpretive coherence, and situational licensing.

1.1 THE PROBLEM OF STRUCTURAL VIABILITY

The first challenge in defining grammaticality is distinguishing between structures that fail because they are fundamentally structurally unsound and those that fail for other reasons. Chomsky (1957), building on formal production systems developed by Post (1943), treated grammaticality as a categorical property defined by membership in a set of well-formed strings. This approach captures the intuition that some strings, like (1a), resist structural analysis entirely:

(2) **Can the have running?*

Here, the input crashes the structural analysis independent of meaning. The “categorical” view correctly identifies that a viable structural analysis is necessary for grammaticality. But by elevating this single condition to the *definition* of grammaticality, formal approaches struggled with evidence that speakers consistently provide gradient judgments for structures that are clearly analyzable but degraded. The competence-performance distinction (Chomsky, 1965) attempted to preserve categorical grammar by attributing gradience to processing limitations. But as Schütze (2016, p. 71) notes, this often leads to circularity, where problematic results are dismissed as performance artifacts. The OVMG framework resolves this by recognizing that while structural viability (the successful mapping of form to category) is a necessary, categorical prerequisite, it isn’t the sole determinant of grammaticality.

1.2 THE PROBLEM OF INTERPRETIVE COHERENCE

The second tension arises from the interplay of form and meaning. Chomsky’s famous example (1b) shows that operator-valued form–value relations can be intact even when compositional semantics fails: the sentence is structurally a declarative, the subject is correctly identified both positionally and through agreement, and the tense suggests a general claim. What breaks is the compositional semantic relation: the lexical values don’t cohere. Conversely, (1c) **I’ve finished it yesterday* shows compositional semantics can be transparent (the intended meaning is clear) while an operator-valued form–value relation fails. Here, the clash is between the present perfect’s temporal value and the adverb’s deictic anchoring.

Generative semanticists like Lakoff (1971) and McCawley (1968) demonstrated early on that many constraints have semantic motivations. Construction Grammar (Goldberg, 1995) has further shown how constructions carry meanings that interact with lexical semantics. Novel uses that align with established constructional meanings (like *She texted him the address*) are accepted, while those that clash (like **She disappeared him the evidence*) are rejected. A complete theory therefore needs to account for interpretive coherence: the degree to which a structurally viable form yields a stable, non-contradictory interpretation.

1.3 THE PROBLEM OF SITUATIONAL LICENSING

Finally, even structurally viable and semantically coherent forms may be ungrammatical if they violate community conventions. Sociolinguistic research (Labov, 1972) demonstrates that grammaticality references community norms. Cross-linguistic variation shows that each language community conventionalizes particular form–value mappings through historical processes. These conventions become entrenched through statistical preemption (Goldberg, 2011) – speakers learn that certain forms are ungrammatical precisely because they encounter alternative expressions in contexts where the ungrammatical form would otherwise be expected.

Consider (1f), repeated here:

(3) A: *How old are you?* B: **I have 25 years.*

This utterance is structurally viable and semantically transparent (the intended meaning is clear), but it is categorically rejected in English. The same form is grammatical in French (*J’ai 25 ans*) and

Spanish (*Tengo 25 años*), demonstrating that the constraint is community-specific. Usage-based approaches (J. Bybee, 2006) emphasize that such rejection arises from the absence of situational licensing. The English-speaking community has conventionalized a different way to express age, leaving this form without positive evidence.

These three problems – structural viability, interpretive coherence, and situational licensing – correspond exactly to the three constitutive variables of the OVMG framework presented in Section 2. Acknowledging them as distinct but interacting components makes it possible to move beyond the impasse of trying to force all grammaticality judgments into a single “competence” definition or a purely probabilistic usage model.

2 THE OPERATOR-VALUE MODEL OF GRAMMATICALITY

This section presents the proposed OVMG framework. The term **VALUE** is used here in the Saussurean sense of *valeur* (Saussure, 1916): the identity of a linguistic unit is constituted not by intrinsic properties but by its position in a system of contrasts, what it patterns with, what it opposes, and what interpretations it makes available. It is not to be confused with feature-value assignment or decision-theoretic value. Throughout, I treat grammatical knowledge as a conditioned form–value relation; when this relation is sufficiently stable in a communicative situation, with a dominant value reliably recoverable and socially licensed, I refer to it informally as an established form–value relationship.

The model’s name marks its unit: operator values – closed-paradigm, update-configuring contrasts – rather than any particular expressive substance. Tone, particles, word order, inflection, and suprasegmental material all fall within its scope when they realize an operator contrast (Reynolds, 2026c).

The formal architecture rests on three constitutive variables: mapping viability (map), operator-value coherence (K), and situational licensing (C_t). Together they specify a **PROFILE** in the projectibility-first sense: a pattern whose partial observation licenses further expectations (Reynolds, 2026a). Their interaction determines grammaticality (§3); what the profile projects, and what supports those projections, is the business of §3 and §4. Figure 1 lays out this order of explanation, which the rest of the paper fills in. Table 1 summarizes these variables; the formal definitions appear in §3.

Variable	Component
$\text{map}(u, c)$	Mapping Viability (Categorical prerequisite)
$K(u, c)$	Operator-Value Coherence
$C_t(u, c)$	Situational Licensing (Social convention)

Table 1: The three constitutive variables of the OVMG framework.

For expository purposes, it is useful to distinguish recurrent **INSTABILITY MODES**: diagnostic categories that correspond to different ways of driving map , K , or C_t toward zero, plus processing and prescriptive factors that modulate how grammatical something *feels* without affecting grammatical status directly.

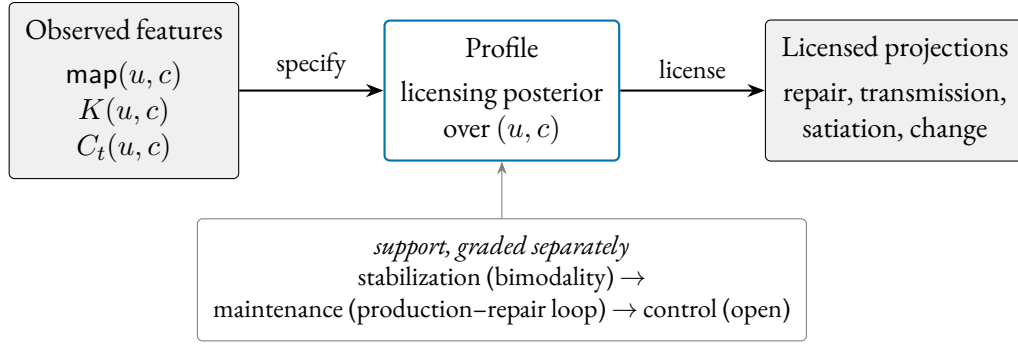


Figure 1: The projectibility-first order of explanation. The three constitutive variables specify a profile – the population licensing posterior – whose partial observation licenses the projections on the right. What stabilizes, maintains, or controls the profile is a separate, graded claim (below), not something inherited from naming the mechanism.

2.1 DIAGNOSTIC CATEGORIES

The diagnostic categories described below aren’t primitives of the model but rather convenient labels for the most common routes to ungrammaticality. Each can be traced to one or more of the constitutive variables, as the formal core in §3 makes explicit.

2.1.1 Operator-valued form–value relations within communicative situations

At its foundation, grammaticality depends on stable relationships between linguistic forms and the operator values they realize within specific communicative situations. That form–value relations exist at all levels of grammatical organization, from morphemes to abstract syntactic patterns, is a core insight of Construction Grammar (Goldberg, 1995). OVMG narrows the target of categorical grammaticality to role-functional contrasts: closed-paradigm, high-opportunity, update-critical resources that configure how an utterance enters public coordination. A companion paper develops this operator-stratum account in detail and argues that the boundary is functional rather than substance-based (Reynolds, 2026c).

Operator exponents, like words, are typically polysemous: a single form participates in multiple form–value relationships, usually with one dominant value (Kilgariff, 2004). The English past tense, for instance, usually denotes past time, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow...*). What matters for grammaticality is whether a stable relationship exists within the relevant communicative situation, not whether the form has a single fixed meaning.

Neuroimaging work provides converging evidence for this integration of form and meaning at the processing level. Fedorenko and colleagues find that brain regions activated by syntactic manipulations overlap substantially with those activated by semantic ones (Fedorenko et al., 2011, 2012, 2024), suggesting partly shared processing resources rather than wholly separate modules.

Following Wiese (2023, p. 3), grammaticality is conditioned by regularities within a “communicative situation”. The emergence claim defended below is narrower: categoricity emerges from the licensing dynamics in §4.4, not from an unanalyzed appeal to usage.

Communicative situations aren’t static or mutually exclusive. A speaker navigates overlapping contexts that shift on multiple dimensions: the immediate situation (formal meeting vs. casual chat), durable social positions (regional, generational), and the norms they’re orienting to (which can shift

with audience). This fluidity explains style-shifting, dialect formation, and the gradual acceptance of initially marginal constructions. What matters for grammaticality is the stability of form–value relationships within the relevant communicative situation.

This is visible in early child language, where toddlers produce utterances that deviate from adult norms but remain internally consistent within the child’s developing system. A toddler in a monolingual English household might say:

- (4) *Ava cookie.* (intended as ‘Ava = I want a cookie’)

Although this differs from adult English norms, it may not be perceived as ungrammatical by the child’s regular caregivers. Used with the same pragmasemantic force among anglophone adults, the construction would be judged ungrammatical.

Multiple modal constructions provide another clear example of variety-relative grammaticality:

- (5) *I might could help you with that.*

This combination of modal auxiliaries is systematically possible for some American English speakers, who can productively generate similar constructions (Morin & Grieve, 2024). Speakers from communities where only single modals are grammatical, though, typically reject such combinations as ungrammatical.

A similar dynamic appears in code-mixing among bilingual speakers, where combinations of forms from different languages can be grammatical within that bilingual community’s norms but not without. Consider a Spanish-English bilingual speaker who uses a Spanish progressive auxiliary with an English lexical verb:

- (6) *Ayer, estábamos lifting en el gym durante una hora.*
 yesterday be.IMPF-1PL lifting in the gym for an hour
 ‘Yesterday, we were lifting in the gym for an hour.’

Within the right communicative situation, this utterance is grammatical. The Spanish auxiliary *estábamos* combines with an English participial form *lifting* to form the progressive aspect. This cross-linguistic combination of morphology and a lexical verb is consistent with local norms, where code-mixed utterances are common and meaningful. In contrast, a standard monolingual Spanish community, which expects fully Spanish progressive structures (*estábamos levantando pesas*), may judge the example in (6) as ungrammatical. The use of intransitive *lifting*, specific to the gym community, further illustrates just how localized grammaticality judgments can be.

This doesn’t mean bilingual communities simply accept any combination of languages. As Toribio (2001) reports, Spanish–English bilingual speakers judge examples like (7) as unacceptable, showing that even in bilingual communities, there are systematic constraints on which language combinations are permitted.

- (7) * *Los enanitos intentaron pero no succeeded in awakening Snow White* (Toribio, 2001)
 the dwarfs try.PST-3PL but not succeeded in awakening Snow White
 ‘The dwarfs tried but did not succeed in awakening Snow White.’

In a slightly different case, as a second-language speaker of Japanese, I used to say

- (8) **Kawaii da.*
 cute-PRES COP.PRES
 ‘(That)’s cute.’ (intended)

Conventionally, though, the redundant tense marking has no accepted meaning, making the use of the copula ungrammatical to most Japanese speakers, though it felt meaningful and grammatical to me. The example underscores that stable form–value relations emerge from and depend on the shared linguistic routines of a particular community, and individuals with differing trajectories of acquisition may diverge in their grammatical judgments.

The specific linguistic context can matter too:

To take an obvious case which Jerry Morgan [(1973)] discussed recently, certain combinations of words are extremely strange if presented in isolation but are perfectly normal as answers to certain questions. *Spiro conjectures Ex-Lax* would generally be felt to be unintelligible if presented out of context but is a perfectly normal answer to the question *Does anyone know what Mrs. Nixon frosts her cakes with?* (McCawley, 1974, 252, italics added)¹

These examples illustrate that grammaticality is communicative-situation-relative. Conversants who differ in immediate situation, social position, or the norms they orient to may disagree on whether a form is grammatical – and both may be right within their respective contexts. The linguist’s task is therefore to determine whether a form is grammatical from any position in a communicative situation, or systematically excluded from all.

The stability of form–value relations within speech communities can be empirically modeled, as demonstrated by Blythe and Croft (2009). Their utterance selection model treats speech communities as networks where speakers track and reproduce linguistic variants based on their interactions. When applied to dialect formation, these models show how competing linguistic variants spread and eventually stabilize, with initial variant frequency strongly predicting which form will prevail.

Different language communities encode different distinctions as grammatically obligatory, reflecting what each community deems relevant enough to systematically mark. Over time, communities establish patterns – grammatical constructions – that reliably signal these chosen distinctions. As a result, what counts as a grammatical necessity in one language may be optional or absent in another.

The progressive aspect provides a useful example. In Standard English, it’s not merely an option but an obligatory grammatical marker for ongoing, incomplete actions:

- (9) *She is studying right now.*

Here, the progressive form *is studying* isn’t just a stylistic choice. It’s the recognized, grammatical way to express a currently unfolding activity. English speakers strongly prefer (and in many contexts, demand) the progressive construction to convey immediacy and ongoingness.

French, by contrast, doesn’t treat the progressive aspect as a grammatically mandatory distinction. While one can signal ongoing activity through adverbs or periphrastic constructions, standard French doesn’t have a dedicated progressive form. The sentence:

¹The humour derives from political tensions of the Watergate era, when both Vice-President Spiro Agnew and President Nixon would ultimately resign from office.

- (10) *Elle étudie maintenant.*
 ‘she studies/is studying now’

can comfortably describe a currently ongoing action without any need for special morphology. The community hasn’t defined this aspectual distinction as an obligatory operator contrast. What is grammatically necessary in English – employing the progressive to signal ongoingness – is simply not a requirement in French.

A similar dynamic appears with evidentiality, the grammatical marking of information sources. In Turkish, evidentiality is systematically encoded through verb forms and particles that distinguish between directly witnessed events and those inferred or reported:

- (11) *Gelmiş*
 ‘He/she came (apparently)’
 (i.e. the speaker wasn’t a witness but inferred or heard about it.)

The language community treats evidential distinctions as central enough to be baked into the grammar. A speaker can’t simply omit evidential marking without sounding ungrammatical.

Of course, while English speakers can say *I heard that he arrived* or *He must have arrived*, these are optional lexical or modal resources rather than required elements of the grammar. The English community simply doesn’t regard evidential distinctions as something that has to be encoded as an obligatory operator contrast.

Kilani-Schoch and Dressler (2005) demonstrate this principle systematically in their analysis of verbal morphology across French and other Romance languages. They show how apparently similar verbal systems can encode quite different semantic distinctions as grammatically obligatory, reflecting each community’s conventions about which meaning distinctions have to be systematically marked. For instance, while both French and Italian mark aspect morphologically, they differ in which aspectual distinctions are grammaticalized versus left to optional lexical expression.

Establishing a form–value mapping is a prerequisite for grammaticality, but it isn’t sufficient. Individuals routinely interpret strings that are nevertheless judged ill-formed within the relevant communicative situation. That very flexibility forces the grammar to police *how* a relation is licensed, not merely *whether* one can be imagined in principle. In other words, additional factors accumulate evidence against candidate mappings: (i) conflict with conventional constructional values contributes negative evidence (§2.1.2), (ii) high recovery cost down-weights the mapping (§2.1.3), and (iii) absence of conventionalization leaves the mapping with insufficient positive evidence (§2.1.6). Only by integrating these weighted contributions does the framework predict the empirical fact that **Furiously sleep ideas green colorless* elicits universal rejection while (*ie*) feels bad in everyday quotation yet is accepted as “grammatical but hard to process” once speakers are walked through the intended structure. The combined weighting therefore concentrates probability mass on the much smaller subset of imaginable relations recognized as bona fide grammatical resources.

One might object that the community’s verdict alone should settle the matter: if speakers converge on using a form, why not define grammaticality as simply “whatever the community accepts”? That move is circular and empirically empty: it can neither flag incipient innovations whose status is still contested nor explain why stable dialectal differences persist despite mutual intelligibility. A purely sociological definition would also mis-classify well-attested performance errors (e.g. agreement slips in live speech) as “grammatical” whenever they pass unnoticed, and it would leave us

with no basis for diagnosing why learners systematically avoid forms that the target community fully endorses. Separating the mapping, compatibility, and acceptance layers therefore avoids trivializing the concept of grammaticality while still granting the community the final say on which relationships endure.

2.1.2 Operator-value compatibility

Beyond the existence of form–value relations, grammaticality requires compatibility among the operator values those relations encode. This component captures cases where individual elements are well formed but their combination creates conflicts among grammaticalized contrasts activated in the communicative situation.

Consider the temporal incompatibility in:

(12) * *I've finished it yesterday.*

Here, the present perfect construction encodes current relevance while *yesterday* specifies completed past time. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows a direct conflict between the temporal value encoded by the grammatical construction and the lexical temporal specification.

The age example belongs elsewhere:

(13) * *I have 25 years.* intended as 'I'm 25 years old'

In English, *have+years* is a licensed frame for duration, remaining time, or experience, but ordinary age predication is licensed through a different frame. The intended value is transparent, so this is not a K failure in the narrow sense. It is a licensing failure for an age-predication construction in English, even though cognate-looking frames are licensed in French and Spanish.

Sometimes value incompatibility involves conflicting information-structural requirements:

(14) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

Here there is a clash in information structure: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit K . The live ϕ_j in the energy model are constraints on grammaticalized, operator-like features in c : tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-structural requirements where a construction grammatically packages focus or backgrounding, and indexical contrasts only when they have entered a closed accountable repertoire. This makes K partly derivative of licensing facts at the feature level. That dependence is a virtue, because the model should not treat arbitrary lexical implausibility as grammatical incoherence.

2.1.3 Processing constraints

Language processing engages both dedicated language networks and domain-general cognitive systems (Fedorenko et al., 2024). When constructions overload these systems – particularly through multiple long-distance dependencies or heavy embedding – they may trigger feelings of ungrammaticality despite being structurally well-formed.

(15) *The bread the baker the apprentice helped made is delicious.*

Evidence for these constraints comes from multiple sources. Studies of sentence processing show that dependencies spanning multiple intervening elements increase processing difficulty, with new referents between dependent elements compounding memory load (Gibson, 2000, 2026). In (15), while each individual relation (like *the apprentice helped* and *the baker made*) is interpretable in isolation, their nested combination overwhelms incremental processing. The construction is grammatical, but excessive bridging costs from its nested structure trigger feelings of ungrammaticality.

Rather than reflecting simple memory limitations, processing constraints emerge from the interaction between specialized language networks and other cognitive systems (Fedorenko et al., 2024). On this processing-level interpretation, what speakers experience as difficulty may reflect demands distributed across specialized brain systems rather than a single cognitive bottleneck.

The diachronic claim requires a further bridge through speaker behaviour and transmission. Forms that impose lower processing demands are easier to store, recall, and reuse. When speakers preferentially select these more manageable structures, simpler patterns can spread through the linguistic community and eventually become conventional. This process can be seen at work when languages simplify nested relative clauses or reduce complex morphological paradigms. The processing evidence therefore supports a plausibility claim about one pathway into conventional stability, not a direct neural explanation of grammatical change.

Processing constraints also interact with other components. A construction that is marginal due to low situational acceptance may become completely unacceptable when combined with processing difficulty. Conversely, highly entrenched constructions may remain acceptable despite considerable processing demands, suggesting that strong conventionalization can partially offset processing costs.

2.1.4 Dependency locality

Integration cost rises with dependency length; self-paced reading shows difficulty peaks at loci of long-distance integration, with locality accounting for 59–73% of variance in reading times (Grodner & Gibson, 2005). Because locality effects are so robust and well-documented, the formal treatment in §3 treats dependency locality as a dedicated component rather than hiding it inside a residual term.

Locality influences both the subjective feeling of ungrammaticality and the licensing dynamics. High-locality constructions feel worse to process, even when structurally well-formed. And because such constructions are harder to produce and parse, speakers tend to avoid them, which over time reduces their conventionalization. The formal treatment appears in §3.

2.1.5 Socio-pragmatic indexicality

The value of a construction extends beyond semantic content to include socio-pragmatic dimensions – how linguistic forms index aspects of social context, speaker identity, group membership, stance, or interpersonal relationships (Eckert, 2012; Silverstein, 1976).

In many varieties of Latin American Spanish, for example (e.g. the Río de la Plata region), speakers use *vos* and its associated verb forms instead of the *tú* forms used elsewhere in the Spanish-speaking world (Bertolotti, 2016):

- (16) ¿*Vos querés un café?*
 you.SG want.2SG-VOS a coffee
 ‘Do you want a coffee?’

Here, the use of *vos* rather than *tú* not only denotes the second-person singular hearer – the person being offered a coffee – but also indexes the speaker’s regional identity and familiarity with the

local dialect. This indexical value may convey closeness, solidarity, or membership in a particular geographic and social community.

This situational view of grammaticality can manifest asymmetrically. Speakers from the Río de la Plata region may view a conversational situation as accommodating both their own norms and those of *tú*-using interlocutors – the situation itself can encompass both *vos* and *tú* as grammatical options. But speakers from *tú*-only regions might conceptualize the same situation more restrictively, defining it in a way that categorically excludes *vos* as a grammatical possibility. This asymmetry doesn't depend on different understandings of the forms themselves, but can arise from different ways of defining what the communicative situation allows, influenced by the indexical values attached to *vos* versus *tú* in their respective communities.

The indexical value of constructions extends to phonology, but as long as there is no conflict with operator value, grammaticality isn't at question. Babel (2025) provides an example. In a study conducted in Bolivia, participants were presented with audio stimuli where only the vowels were manipulated to reflect either a highland or lowland accent. This presented participants with incongruent identity cues (e.g. vowels from one accent along with consonants from the other). But this didn't trigger feelings of ungrammaticality. Instead, vowel contrasts activated expectations about consonant features and discourse markers, resulting in some participants' "hallucinating" identity-linked features that weren't present in the signal.

This demonstrates that the value of a construction goes beyond semantic features to include socio-pragmatic aspects that shape how speakers and hearers negotiate authority, identity, solidarity, and other interpersonal relations. Recognizing these indexical dimensions is essential for understanding why certain forms feel natural and grammatical to some speakers and out of place or even ungrammatical to others.

2.1.6 Situational entrenchment and acceptance

Even well-formed constructions with clear meanings may be judged ungrammatical if they lack conventionalized acceptance. This component captures the degree to which a form–value relation has become entrenched within relevant communicative situations.

(17) * *We sheared three sheeps.*

The regular plural marking is semantically transparent and structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational entrenchment operates independently from the other components. A construction may have perfect value compatibility and minimal processing demands but still be rejected because a different form has been conventionalized for that meaning. This is particularly evident in cases of extreme rarity. Some constructions are so infrequent that speakers lack a shared consensus about their status.

The independent relative genitive pronoun *whose* provides an example, being so unusual that Hankamer and Postal (1973) deem it non-existent. The contexts that license this construction require the simultaneous convergence of distinct pragmatic and syntactic conditions – sufficient accessibil-

ity of both possessor and possessum, the appropriate information structure, and an environment allowing ellipsis – which are seldom met all at once (Reynolds, 2024a):

(18) ? *I saw Joan, a friend of whose was visiting.*

(adapted from Huddleston & Pullum 2002, p. 472)

A search of the 1-billion-word Corpus of Contemporary American English returned no instances of this construction. Crucially, though, the opportunity set is also tiny: contexts requiring an independent relative genitive are vanishingly rare. Speakers therefore have little evidence either way – neither positive tokens nor the systematic absence that would accumulate if the construction were regularly passed over. The result is high *uncertainty* rather than confident rejection. Some speakers can make the analogical leap from similar constructions to accept examples like (18); others remain unsure; still others, lacking sufficient exposure, can't construct a stable form–value relation at all. Even among those who grasp the construction analytically, Shain et al. (2020)'s fMRI research suggests that its unexpectedness would trigger high surprisal, compounding the uncertainty with processing difficulty.

This contrasts sharply with stable gaps like left-branch extraction (§2.1.7), where the opportunity set is large and the construction is systematically untaken – giving speakers overwhelming evidence that it isn't licensed.

2.1.7 Apparent categorical gaps

Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

(19) * *Which did you buy* [___ *car*]?

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of “ungrammaticality”.

Under OVMG, such patterns are analyzed not as hard representational bans but as stable gaps: situational licensing is driven toward zero by persistent preemption. Because the communicative job is common and an established competitor always wins (e.g., *Which car did you buy?*), speakers accumulate overwhelming evidence that the construction isn't part of the repertoire. Additional processing penalties – such as systematic garden-path reanalysis when the bare *wh*-phrase is encountered – can strengthen the subjective categoricity without any independent structural veto being posited. The formal dynamics appear in §3.

Cross-linguistic comparison doesn't threaten this analysis, because the dynamics are defined over a language's own niches. LEFT BRANCH is a descriptive class tied to a particular phrase-structure analysis, not a comparative concept (Haspelmath, 2010); whether discontinuous nominals in case-rich, article-less languages instantiate the same configuration as the English gap is far from clear, and the companion corpus study deliberately brackets the comparison (Reynolds, 2026d). What the model inherits instead is a prediction about opportunity structure: superficially similar discontinuous-nominal patterns should be learnable where inflectional marking makes the dependency recoverable, a productive source of discontinuous order supplies scaffolding, and a discourse niche makes the separated

element useful. English meets none of the three, so the candidate has no scaffolding of its own and the contiguity-preserving competitors win every opportunity.

Several diagnostic criteria help identify these stable-gap constructions:

1. Persistent unacceptability: The construction never gains acceptance; familiarity and repetition don't shift judgments.
2. Categorical rejection: Speakers find the constructions simply impossible, with no intermediate ratings.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn't improve acceptability.

Prediction: If participants are led to treat presented tokens (like LBE) as intentional dialectal or ingroup resources rather than errors, then satiation (or at least reduced feeling of ungrammaticality) should be more likely than in standard paradigms where the construction is treated as simply wrong.

If future work uncovers constructions that resist even this stable-gap analysis – forms for which no structural analysis is available at all – they would represent a residual class of hard representational bans. For now, treating classic cases like LBE as driven to zero under preemption keeps the constitutive core minimal and pushes the explanatory work into the dynamics of situational licensing.

2.1.8 Competitor-less gaps: paradigm defectiveness

Preemption-driven stable gaps have a winning competitor. A different gap species doesn't. Paradigm defectiveness leaves a high-opportunity cell empty without any single form winning it: most English varieties lack a negated contracted form for *am* (**amn't*, though Irish and Scottish English license it), and Russian speakers avoid the expected first-person singular non-past of *pobedit'* 'win', resorting to periphrasis instead. The niche is common and the candidate forms are transparent, yet no variant accumulates the positive evidence to win, plausibly because conflicting analogical pressures split the candidate set. In the present terms this is a third licensing profile: high opportunity with the evidence streams divided among candidates, leaving each candidate's posterior low in mean but unstable, rather than concentrated near zero. The phenomenology matches. Speakers report not knowing how to say it – ineffability – rather than the confident "that's wrong" of a preempted gap. The satiation-framing prediction extends accordingly: defective cells should behave like low-evidence forms, movable under framing, not like stable gaps.

2.2 DIAGNOSING (UN)GRAMMATICALITY AT A GLANCE

For exposition the recurrent instability modes can be read as a short decision tree (the formal definitions appear in §3):

Condition	→ Canonical outcome
no viable structural analysis	→ NONSENSE (* <i>Can the have running</i>)
semantic/indexical clash	→ value incompatibility (* <i>I've finished it yesterday</i>)
rarely encountered, low certainty	→ community-novel (<i>friend of whose</i>)
systematically preempted	→ stable gap (left-branch extraction)
high opportunity, candidates split	→ defective cell (* <i>amn't</i>)
high processing cost, structure recoverable	→ transient ill-formedness (centre embedding)
otherwise	→ grammatical

2.3 PATTERNS OF (UN)GRAMMATICALITY

When these recurring components interact, they produce systematic patterns in how constructions succeed or fail to achieve grammatical status. Understanding these patterns helps explain the diverse phenomena traditionally grouped under “ungrammaticality”

2.3.1 When constructions are ungrammatical

Ungrammaticality arises when expected patterns within the relevant communicative situation are violated in ways that can’t be reconciled through available interpretive strategies. The diagnostic categories generate distinct types of violation:

NO VIABLE FORM–VALUE RELATION. In some cases, the form of an utterance simply doesn’t map onto any conceivable interpretation:

(20) **Can the have running?*

Here, the modal *can* expects a subject and a verb phrase to form a coherent proposition. Instead, *the have running* neither yields a noun phrase nor a legitimate verbal structure. The result is a form for which no stable meaning is available. With no recognizable pattern to anchor on, the utterance remains nonsensical and ungrammatical.

VALUE INCOMPATIBILITY. When the operator values encoded by grammaticalized elements clash, the construction fails despite having well-formed individual components. This includes temporal conflicts (as in **I’ve finished it yesterday*) and information-structural contradictions (as in the lifeguard example).

PROCESSING OVERLOAD. Constructions that exceed cognitive processing capacity trigger ungrammaticality judgments even when structurally well-formed. Multiple center embeddings exemplify this pattern, where incremental processing becomes impossible despite theoretical grammaticality.

LACK OF SITUATIONAL ENTRENCHMENT. Even transparent, processable constructions fail when they violate established conventions. The *sheeps* example shows how regularization can be preempted by entrenched irregular forms; **I have 25 years* fails as ordinary English age predication because the relevant frame is not licensed for that communicative job. The extreme rarity of independent relative *whose* prevents stable conventionalization.

APPARENT CATEGORICAL GAPS. Some constructions are rejected so persistently that they behave as if categorically banned. Under OVMG, these are analyzed as **STABLE GAPS**: the structure is viable and interpretation coherent, but situational licensing is driven toward zero by persistent preemption in a large opportunity set (§4.3, §4.5). Left-branch extraction is the textbook case.

2.3.2 Degrees of violation

Grammaticality isn’t binary; rather, it reflects the scope and intensity of component mismatches. Violations can range from mild, easily recoverable deviations to complete breakdowns in interpretability.

Minor violations might involve a single component with partial conflict – for instance, a construction with moderate processing difficulty but full value compatibility and strong situational entrenchment. Such cases often receive intermediate acceptability ratings and may improve with exposure.

Severe violations typically involve multiple components or complete failure of a single critical component. A construction lacking any viable form–value relation represents total breakdown, while violations of categorical constraints produce consistent, strong rejection regardless of other factors.

The interaction between components can amplify or mitigate violations. Strong situational entrenchment can partially offset processing difficulty, while semantic transparency can make marginal syntactic patterns more acceptable. Conversely, multiple minor violations can compound to produce strong ungrammaticality judgments.

2.4 THE SUBJECTIVE EXPERIENCE OF GRAMMATICALITY

While the constitutive variables determine conventional grammatical status (§3.3), speakers experience violations through subjective feelings and judgments that don't always align perfectly with that status.²

2.4.1 The feeling of ungrammaticality

The subjective **FEELING OF UNGRAMMATICALITY** is speakers' metacognitive response to linguistic violations. This distinction parallels other well-studied metacognitive feelings like the **FEELING OF KNOWING** (FOK; Hart, 1965), which arises when one feels certain of knowing something but can't retrieve it.

Note that there is no positive feeling of grammaticality, just as there is no feeling of having sufficient oxygen, only negative feelings experienced in the absences. The feeling of ungrammaticality, then, can be seen as the negative response triggered when an utterance violates expected form–value patterns.

This notion echoes Edward Sapir's concept of “form-feeling” – an often unconscious grasp of language patterns (Sapir, 1921, 1927). Evidence from aphasia supports the distinction between grammaticality and its subjective experience. Patients with Broca's aphasia, who produce agrammatic speech, often exhibit self-monitoring behavior, attempting self-correction and expressing frustration with their grammatical errors (Oomen et al., 2005).

The feeling of ungrammaticality can be conceptualized as a negative response speakers experience when unstable or missing form–value relations are detected during processing. This response wouldn't define grammaticality itself but would instead serve as a speaker-level heuristic for detection. This distinction helps explain:

1. **Gradient Judgments:** Varying degrees of certainty about marginal constructions reflect differences in the strength of the negative response rather than categorical grammatical status.
2. **Satiation Effects:** Repeated exposure to marginal constructions might not change grammatical status but could attenuate negative responses by increasing familiarity.
3. **Cross-Linguistic Variation:** Differences in how reliably operator-value violations are detected affect the consistency and intensity of negative responses.
4. **Mismatch Between Intuition and Reality:** Constructions can be objectively grammatical but trigger negative responses due to processing difficulty, or vice versa. That is (un)grammaticality can be illusory (Fanselow, 2021).

²The distinction between objective grammaticality and the subjective feeling has a philosophical analogue in Cavell's (1969) distinction between knowledge and acknowledgment. For Cavell, acknowledgment isn't a separate capacity from knowing but an inflection of it – bearing knowledge toward the situation that produced it. Speakers don't merely *know* that a string is ungrammatical; they *register* it as a violation, with metacognitive consequences. This matters for LLM-based grammaticality research: LLMs can produce judgments that correlate with human ones, but whether they acknowledge the norms they're applying – or merely process patterns that correlate with those norms – remains an open question (Techio, 2026).

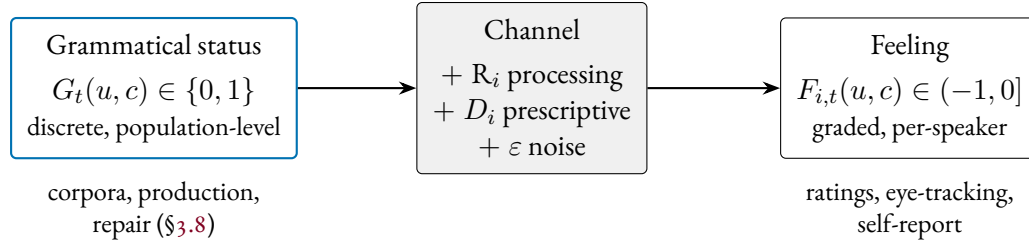


Figure 2: The two-layer architecture. Objective, discrete grammatical status is not observed directly; it is filtered through processing costs, prescriptive dissonance, and noise into the gradient feeling that ratings measure. The two layers are identified by different evidence streams, which is what lets them dissociate and so falsify each other (§3.6).

2.4.2 Distinguishing objective grammaticality from subjective ratings

A fundamental challenge in grammaticality research lies in distinguishing the objective property of grammaticality from the subjective feelings and ratings that speakers provide. This distinction proves crucial for both theoretical development and empirical investigation.

TWO LEVELS OF ANALYSIS.

The framework distinguishes two conceptually distinct levels:

Table 2: Objective grammaticality versus subjective experience

Level	Nature	Observable through
Objective grammaticality	Whether a form–value relation is licensed in the communicative situation	Converging evidence: corpora, production, repair behaviour
Subjective feeling	Metacognitive warning signal that “something is wrong here”	Directly observable: ratings, eye-tracking, self-reports

This distinction parallels the measurement problem in other sciences: temperature can’t be directly observed but has to be inferred through the behaviour of thermometers. Similarly, grammaticality itself remains a theoretical construct accessed through various behavioural measures, with acceptability ratings being just one imperfect window into the underlying phenomenon. Figure 2 shows the two layers: a discrete grammatical status, filtered through processing costs, prescriptive overlays, and noise, yields the graded feeling that ratings actually record.

WHAT FACTOR ANALYSIS REVEALS.

When researchers conduct factor analyses on acceptability ratings – a common approach in experimental syntax – they necessarily target the structure of the subjective feeling, not objective grammaticality directly. The factors that emerge from such analyses reflect both genuine grammatical constraints and additional sources of variance that modulate subjective responses without affecting grammatical status.

- Operator licensing: When this factor loads heavily on rejected items and converges with corpus-rate-per-opportunity, elicited production, and repair behaviour, it reflects C_t .
- Value coherence: This factor, including information-structure clashes, belongs to K only when it tracks constraints on grammaticalized operator values.
- Situational entrenchment/frequency: High-frequency patterns that speakers have conventionalized provide evidence about C_t , but raw frequency has to be normalized by opportunity set.
- Sociolinguistic appropriateness: This factor has mixed status. Indexical clashes can affect C_t when they enter a closed accountable repertoire, but stylistic infelicities typically affect F .
- Processing cost/memory load: This factor belongs to F unless it feeds back into production, repair, and later licensing through the loop in §3.3.2.
- Morphophonological well-formedness: When phonological patterns realize obligatory operator exponents, violations affect K or C_t . Otherwise, they influence the subjective response.

This distinction between objective grammaticality and subjective feeling has immediate methodological consequences. If research aims to uncover the grammar itself, investigators should model factors only when they also appear in the independent indicators for C_t and K . Conversely, research into human judgment behaviour should retain the full factor structure, as it is precisely the subjective feeling that predicts how speakers will rate sentences presented out of context.

The distinction also explains certain puzzling phenomena in the experimental literature. Satiation effects may reflect changes in subjective feeling without necessarily altering objective grammaticality. A construction might remain objectively ungrammatical in the community's system while triggering progressively weaker negative responses through familiarization. Similarly, the persistence of gradient judgments for certain phenomena suggests that subjective response operates on a continuum even when the population licensing state is effectively categorical (§4.4).

Most importantly, recognizing this distinction prevents circular reasoning in grammatical theory. Rather than defining grammaticality through acceptability ratings – which would conflate objective status with subjective response – the framework treats ratings as one type of evidence among many for inferring the underlying grammatical system. This approach maintains the empirical vulnerability of grammatical hypotheses while acknowledging the complex relationship between competence and performance, between the linguistic system of a communicative situation and individual speakers' responses to it.

2.4.3 Misattribution effects

Sometimes the subjective feeling of ungrammaticality doesn't accurately reflect objective grammatical status. Both false positives (grammatical constructions feeling ungrammatical) and false negatives (ungrammatical constructions escaping detection) occur systematically.

WHEN GRAMMATICAL CONSTRUCTIONS FEEL UNGRAMMATICAL.

Listeners and readers can misanalyze utterances, triggering feelings of ungrammaticality even though the construction conforms to standard patterns:

(21) *The old man the boats.* (Ritchie & Thompson, 1984)

On first reading, one might treat *old man* as a noun phrase, resulting in nonsense. But the sentence actually has *the old* as the subject (meaning 'the elderly'), and *man* as a verb. Interpreted correctly, the sentence means 'The elderly operate the boats', and is fully grammatical.

Processing overload provides another source of misattribution. Heavily nested structures like

(15) are grammatically well-formed but trigger negative responses due to cognitive limitations rather than grammatical violations.

WHEN UNGRAMMATICAL CONSTRUCTIONS ESCAPE DETECTION.

Conversely, objectively ungrammatical utterances may fail to trigger negative responses when semantic content is compelling:

- (22) * *In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The intended interpretation is clear, and hearers readily infer the meaningful comparison. Syntactically, however, the structure means something else. The first clause suggests comparing numbers of people, but the second clause has *they* ('more people'), making the meaning 'more people did x than more people did y', which is nonsensical. But this violation often escapes detection.

Agreement errors can similarly hide within complex noun phrases:

- (23) * *The patchwork of laws governing background checks, addressing assault-weapons limits, and regulating open-carry practices help explain why people continue to be wounded and killed.*
(modified from Corbett, 2016)

The singular subject *patchwork* clashes with plural *help*, but the intervening plural nouns mask this violation. Cognitive resources devoted to processing complexity can prevent speakers from registering the agreement error.

2.5 MECHANISMS OF GRAMMATICAL CHANGE

The stability of form–value relations isn't static. Various motivations drive the acceptance of new forms and the rejection of established ones, explaining how grammatical systems evolve over time.

2.5.1 Semantic motivations for reanalysis

Speakers regularly deploy metaphors, analogies, and context-induced reinterpretations that stretch or shift the meaning of existing forms. Over time, such re-analyses can yield new grammatical constructions that weren't previously part of the language's repertoire.

A well-documented example is the development of the *going to* futurate construction. Historically, *going to* described physical motion toward a place:

- (24) *I am going to London.*

Frequent usage in contexts where motion implied subsequent action (e.g. *I am going to fetch some firewood*) led to semantic shift. The directional component was reinterpreted as marking future intention rather than spatial goals:

- (25) *It's going to rain.*

What once expressed physical trajectory now functions as a predictive futurate marker. The form–value relation changed through semantic reanalysis, driven by communicative utility.

2.5.2 Social motivations for accepting or rejecting forms

Linguistic variants often serve as markers of regional background, class, ethnic identity, or age group. These social cues motivate speakers to favor some forms over others, leading to shifts in what counts as grammatical.

The decline of the simple past (*le passé simple*) in modern spoken French illustrates social motivation. Forms like *il alla* ('he went') became associated with formal, literary, or provincial speech. To avoid signaling undesirable social affiliations, speakers gravitated toward compound forms like *il est allé*, which lacked these status-laden connotations.

This transformation underlines how grammatical acceptance reflects changing social dynamics. Forms that were once fully grammatical can become socially marked and gradually excluded from normative grammar.

2.5.3 Structural motivations

Structural considerations arise from cognitive and communicative demands, including processing limitations, the need for clarity, and the influence of analogy.

PROCESSING CONSTRAINTS AND MEMORABILITY. Languages favor patterns that minimize processing demands. Forms requiring multiple long-distance dependencies or heavy embedding are less likely to achieve stable transmission across generations. This pressure toward processability shapes grammatical evolution, with simpler patterns spreading through communities.

ANALOGICAL EXTENSION. When speakers encounter new communicative challenges, they often extend known patterns to new contexts:

(26) [?] *I asked about what did she do.*

Though currently marginal, such forms appear to gain traction by analogy with main-clause interrogatives (*What did she do?*). If accepted, this analogical extension would represent structural motivation reshaping the grammar.

2.5.4 Iconic motivations

Forms are sometimes accepted precisely because their structure reflects their meaning. Reduplication for intensity provides a simple example:

(27) *a big big problem*

The formal repetition iconically mirrors the magnitude being expressed. Such patterns are readily interpretable and likely to spread because the form–value link is immediately apparent.

Note that iconicity remains syntactically constrained. It occurs in pre-head modifiers but is ungrammatical as predicative complement:

(28) **the problem was big big*

These diverse motivations – semantic, social, structural, and iconic – can interact and sometimes conflict. While Optimality Theory has productively modeled such competing pressures through ranked constraints (Prince & Smolensky, 2004), OVMG views their resolution as arising from usage within communicative situations rather than solely from fixed rankings. The specific weighting of motivations reflects historically established patterns and communicative needs.

3 A FORMAL CORE: GRAMMATICALITY AS CONDITIONED STABILITY

The framework separates two explanatory targets: (i) a **STATE THEORY** specifying what it is for an utterance type to be grammatical *for a population in a communicative situation* at time t ; (ii) a **DYNAMICS MODULE** specifying how such states tend to arise and persist. In earlier vocabulary, the state theory is constitutive and the dynamics module etiological. A more exact phrasing: the state theory specifies the **PROFILE**, and the dynamics module states its **SUPPORT** – what stabilizes the profile and, where the evidence warrants, what maintains it. The phrasing matters because support claims come in grades that have to be earned separately: stability, maintenance, and corrective control are different achievements, and none is inherited from the framework’s name (Reynolds, 2026a, 2026b). Table 3 summarizes the notation.

Table 3: Notation summary

Symbol	Interpretation
<i>State theory</i>	
u	Utterance type or candidate token
$g \in \mathcal{H}$	Construction node in a hierarchy
$c \in \mathcal{C}$	Conditioning state (situation + norm-centre)
$\text{map}(u, c)$	Structural viability ($\in \{0, 1\}$)
$K(u, c)$	Operator-value coherence with reject option ($\in [0, 1]$)
$C_t(u, c)$	Situational licensing at time t ($\in [0, 1]$)
$\tilde{G}_t(u, c)$	Graded stability score ($= \text{map} \cdot K \cdot C_t$)
<i>Feeling model</i>	
$F_{i,t}(u, c)$	Subjective feeling of anomaly
$L(u)$	Dependency locality cost
$R_i(u, c)$	Processing cost vector (interference, locality, surprisal)
<i>Dynamics (§4)</i>	
$N_t(n, c)$	Number of opportunities for niche n in state c
$\rho_t^*(v \mid n, c)$	Counterfactual choice probability among analogical candidates
s_t, e_t, p_t	Positive evidence, error evidence, preemption mass

3.1 CONDITIONING STRUCTURE

Let $c \in \mathcal{C}$ be a **CONDITIONING STATE** – a construed communicative situation (Wiese (2023)) together with whatever norm-centre is treated as relevant. Nothing here requires c to be externally given; interlocutors can misalign about c , and c can be learned, split, and renegotiated over time.

For many purposes it is useful to think of c as a bundle that may include situational features (activity type, medium, footing), stable baselines tied to speaker ascription, and norm-orientation (discourse-community identification), but the formalism treats c abstractly as the conditioning variable that selects the relevant distribution of form–value relations.

3.2 OBJECTS

Let u range over utterance tokens or fine-grained utterance types, with structural representation $M(u)$. Let $\sigma(u, c)$ be the utterance-level interpretation made available in c (lexical content plus prag-

matic enrichment, indexical resolution, discourse update, etc.).

Utterance types inherit licensing from a hierarchy of constructional nodes $\mathcal{H}$. A novel token such as *She texted him the address* need not have been encountered before; it can still be covered by licensed nodes for ditransitive transfer, pronominal object order, and the relevant lexical-class relations. Let $A(u) \subset \mathcal{H}$ be the nodes activated by u , and let $\lambda_g(u, c)$ be non-negative coverage weights with $\sum_{g \in A(u)} \lambda_g(u, c) = 1$. Each node g has its own licensing posterior. A token-level licensing estimate can then be derived by partial pooling:

$$C_t(u, c) = \sum_{g \in A(u)} \lambda_g(u, c) C_t(g, c).$$

The sum is schematic; a similarity kernel over representations would do the same work. What matters is that the learner updates constructional regions, not only memorized sentence types. This prevents the dynamics from predicting that every first-heard sentence has zero licensing.

The framework distinguishes three state quantities.

(I) **MAPPING VIABILITY**. Let $\text{map}(u, c) \in \{0, 1\}$ indicate whether structural analysis can yield at least one well-typed representation for u (category assignment + structural composition) throughout the derivation.

$$\text{map}(u, c) = \begin{cases} 1 & \text{if } \exists \text{ valid analysis } M(u) \text{ yielding } \mu(M) \\ 0 & \text{otherwise.} \end{cases}$$

map is prior to *situational licensing* in one specific sense: it registers whether any covering analysis exists at all (word salad and category errors defeat it) while enforcing nothing about whether the covered analysis is licensed in c . Situational non-licensing belongs only in C_t . But map isn't pre-conventional; §3.4 spells out its dependence on the constructional hierarchy, stored wholes included.

(II) **COHERENCE AS SATISFIED CONCENTRATION OF INTERPRETATION**. Even when $\text{map}(u, c) = 1$, interpretation can be unstable: multiple incompatible construals may compete, or the best available construal may violate an operator constraint. A normalized distribution over ordinary interpretations would measure ambiguity but not violation: a single catastrophic construal would still receive probability 1. To avoid that bug, the model adds a null outcome ω_0 representing rejection or failed operator-value satisfaction. Let $\Omega^+(u, c) = \Omega(u, c) \cup \{\omega_0\}$ and fix $E(\omega_0; u, c) = \kappa_c$:

$$p(\omega \mid u, c) = \frac{e^{-E(\omega; u, c)}}{e^{-\kappa_c} + \sum_{\omega' \in \Omega(u, c)} e^{-E(\omega'; u, c)}}.$$

$$K(u, c) = \max_{\omega \in \Omega(u, c)} p(\omega \mid u, c) \in [0, 1].$$

I choose the null-outcome repair because it preserves the probabilistic interpretation of K . An unnormalized score $e^{-E(\omega^*)}$ or an explicit $K_{\text{peak}} \times K_{\text{satisfaction}}$ decomposition would make similar predictions, but the null outcome keeps ambiguity and satisfaction in a single race. A good single construal has $E(\omega^*) \ll \kappa_c$ and $K \approx 1$; a transparent but operator-violating construal has $E(\omega^*) \gg \kappa_c$, so probability mass shifts to ω_0 and $K \approx 0$.

The energy model $E(\omega; u, c)$ is *open-ended*:

$$E(\omega; u, c) = \sum_{j \in \mathcal{J}(c)} w_j \phi_j(\omega; u, c),$$

where $\mathcal{J}(c)$ includes constraints on grammaticalized features live in c . In the demonstrations below, I assume a minimal inventory: (1) temporal alignment where tense–aspect is grammaticalized, (2) argument-structure satisfaction where a construction licenses roles, (3) information-structural fit where topic/focus packaging is grammatically specified, and (4) indexical consistency only when the contrast belongs to a closed accountable repertoire. This criterion makes K partly derivative of licensing facts. A value can enter E only after the community has conventionalized it as a grammaticalized constraint.

This decomposition bridges the narrative diagnostic categories of §2.1 to the formal variables. For instance, coherence demands may be grouped by their subsystem – denoting the sum over purely semantic demands as E_{sem} and over indexical/pragmatic demands as E_{index} – to yield an expositional factorization $K \approx K_{\text{sem}} \cdot K_{\text{index}}$. In practice, however, the framework assumes that stability depends on the joint concentration of all constraints live in the communicative situation c .

(III) SITUATIONAL LICENSING (ENTRENCHMENT) UNDER CONDITIONING. Each individual i maintains a graded licensing state for u in c : a credence $\pi_{i,t}(u, c) \in [0, 1]$ that u is a community resource there. §4.3 models this state as a Beta posterior. The licensing *attitude* $\Lambda_{i,t}(u, c) \in \{0, 1\}$ – treating the relationship as a legitimate option rather than a performance slip or nonce error – is a thresholded read-out of that graded state, not a separate primitive.

The population-level licensing state is the distribution of $\pi_{i,t}(u, c)$ across the population induced by c . Its mean,

$$C_t(u, c) = \mathbb{E}_i[\pi_{i,t}(u, c)] \in [0, 1],$$

is the object previously called “entrenchment”, explicitly conditioned on c . The population quantified over isn’t a separate primitive; it is induced by the norm-centre and communicative situation bundled in c .

The mean doesn’t exhaust the state. A community in which half the speakers confidently license u and half confidently reject it, and a community in which every speaker sits at credence one-half, share $C_t = 0.5$ but differ in dispersion, both across speakers and within each speaker’s posterior. Stable mixture and shared uncertainty are different conditions with different signatures (bimodal versus unimodal individual judgment distributions, high versus low within-speaker consistency), and §3.3.2 shows that some of the model’s projections run over exactly that difference. Where only the mean matters, I write C_t ; where concentration matters, I refer to the licensing state.

3.3 GRAMMATICAL STATUS AS A CONDITIONED STATE PROPERTY

I write $\mathbb{I}[\cdot]$ for the indicator function: $\mathbb{I}[\phi] = 1$ if ϕ is true and 0 otherwise.

The state theory distinguishes a graded stability score from a categorical membership predicate.

3.3.1 A graded stability score

Recall that $\text{map}(u, c) \in \{0, 1\}$ indicates whether u has at least one available structural analysis in conditioning state c that yields a well-defined operator value. Let $K(u, c) \in [0, 1]$ be coherence as satisfied concentration of the interpretation distribution, and let $C_t(u, c) \in [0, 1]$ be the population licensing rate at time t in c .

Define the graded stability score:

$$\tilde{G}_t(u, c) = \text{map}(u, c) \cdot C_t(u, c) \cdot K(u, c) \in [0, 1].$$

This quantity is the workhorse for modeling gradience and for linking the state theory to behavioural measures.

3.3.2 Projectible use of the graded score

$\tilde{G}_t$ is not bookkeeping after the real explanation has happened. It is a macro-state over which grammaticality projects. But the projections split into two grades, and the split matters.

Some generalizations run over $\tilde{G}_t$ as such. Production and repair behaviour at time $t + 1$ depend partly on $\tilde{G}_t$:

$$\begin{aligned} \Pr(\text{produce } u \mid n, c, t) &\propto \tilde{G}_t(u, c) \rho_t^*(u \mid n, c), \\ \Pr(\text{repair } u \mid c, t) &= r_0 + r_1(1 - \tilde{G}_t(u, c)), \quad r_1 > 0. \end{aligned}$$

For these, a low score licenses the same expectations regardless of whether it comes from mapping failure, operator-value incoherence, or non-licensing, and high- $\tilde{G}$ forms show the converse profile.

Other generalizations need the finer state: the licensing posterior with its concentration, not just its mean. Two constructions can share a low C_t while differing in how much evidence backs it, and the update rule (29) makes them behave differently under new exposure: a diffuse posterior should move with framing and familiarization, while a concentrated one shouldn't. Satiation profiles, framing sensitivity (§2.1.7), and actuation (§4) are therefore projections over the posterior state. This is what separates the independent relative *whose* (low evidence, diffuse posterior) from left-branch extraction (heavy preemption, posterior concentrated near zero): the two can approach similar means while projecting differently.

The feedback loop closes as before. Production and repair generate the later evidence streams s_t , e_t , and p_t , so individual behaviour updates population licensing, and the resulting population state projects back into future production and repair. The projectible generalizations are satiation profiles, transmission fidelity, repair rates, and change trajectories – some over $\tilde{G}_t$ as such, some over the posterior state that $\tilde{G}_t$ summarizes.

Figure 3 makes the two-dimensional state visible. The horizontal axis is the licensing mean, which $\tilde{G}_t$ reads; the vertical axis is the posterior's concentration, which it discards. Stable gaps and defective cells share a low mean and separate only on the vertical axis, so any account that projects from the mean alone cannot tell confident rejection from unresolved uncertainty.

3.3.3 A categorical membership predicate

Communities often treat grammaticality as a constitutive membership fact: either a relationship counts as an available resource in c or it doesn't. A categorical predicate can be defined by thresholding the graded score:

$$G_t(u, c) = \mathbb{I}[\tilde{G}_t(u, c) \geq \tau(c)].$$

The threshold $\tau(c)$ isn't a hidden grammatical parameter. It is a decision criterion associated with the communicative situation: some situations are strict about what counts as “in the repertoire”, others are permissive.

3.3.4 A decision-theoretic motivation

Suppose a judge in c is choosing between two labels, $\ell \in \{\text{IN-REPERTOIRE}, \text{NOT-IN-REPERTOIRE}\}$, and faces asymmetric losses. Let $L_{\text{FA}}(c)$ be the loss of treating an item as IN-REPERTOIRE when it

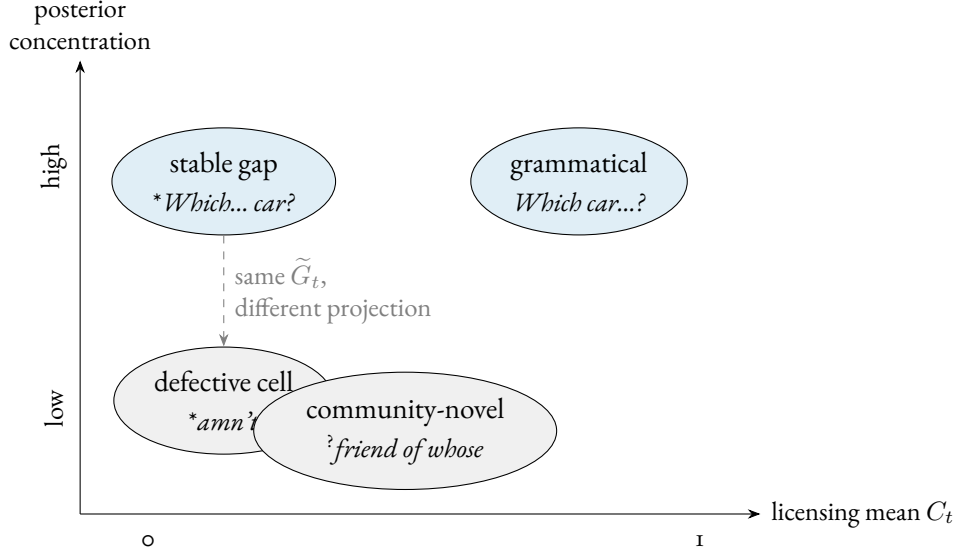


Figure 3: The licensing state has two dimensions, but the mean-product $\tilde{G}_t$ reads only the horizontal one. Stable gaps (dense preemption, posterior concentrated near zero) and defective cells (split candidates, posterior low but diffuse) share a licensing mean and part only on concentration – the axis that predicts satiation, framing sensitivity, and actuation.

isn't, and $L_{\text{FR}}(c)$ the loss of treating an item as NOT-IN-REPERTOIRE when it is. Under the simplifying assumption that $\tilde{G}_t(u, c)$ is a calibrated signal of stability, the optimal rule is to accept u as IN-REPERTOIRE whenever:

$$\tilde{G}_t(u, c) \geq \tau(c) = \frac{L_{\text{FA}}(c)}{L_{\text{FA}}(c) + L_{\text{FR}}(c)}.$$

On this view, $\tau(c)$ is induced by payoff structure: high-stakes institutional contexts raise L_{FA} (and thus τ), whereas in-group, low-stakes contexts raise L_{FR} (lowering τ).

Two clarifications keep the categorical predicate from overclaiming. First, calling the status “objective” means population-level and convention-constituted, not judge-independent in any stronger sense. The status stands apart from any individual’s response the way facts about word meaning or currency do; it is worldly in that, given the community’s practices, the projected relations aren’t made available by the analyst’s decision to group cases together (Reynolds, 2026a). CONVENTIONAL STATUS would be an equally accurate name, and I use the two interchangeably below. Second, the threshold’s arbitrariness is confined by the dynamics: where licensing posteriors are bimodal (§4.4), G_t is insensitive to the exact value of $\tau(c)$ and the categorical predicate does its work; in the contested middle, judgments should be unstable and stakes-sensitive, which is a prediction rather than an embarrassment. The categorical claim is thus a scoped one: it holds where the population state is bimodal and is silent elsewhere.

3.4 WHAT map DOES AND DOESN'T DO

The mapping predicate $\text{map}(u, c)$ is reserved for genuine analyzability failure: cases where no structural analysis is available that yields any well-defined operator value in c . This is the only categorical

failure mode in the representational sense.

Analyzability is relative to the constructional hierarchy $\mathcal{H}$, stored wholes included, not to productive composition alone. Syntactically anomalous idioms make the difference visible: *by and large*, *far be it from me*, and *come Monday* have no analysis by the productive syntax of English, yet they are fully conventional (Fillmore et al., 1988). If map meant “parsable by general rules”, these would come out map = 0 and hence ungrammatical – the wrong result. They are covered instead by their own stored nodes in $\mathcal{H}$: coverage by some licensed node, productive or memorized, is what analyzability means here. The cost of this choice is candour about status: map isn’t pre-conventional. What it contributes is a different *kind* of failure – no covering node at all – not a convention-free one.

Importantly, phenomena traditionally described as categorical “structural bans” aren’t treated here as mapping failures. In those cases, the intended analysis is typically available as a candidate representation – so $\text{map}(u, c) = 1$ – and the intended construal can be coherent once stipulated (so $K(u, c)$ need not be small). What makes the pattern behave categorically in the community is instead the licensing term: under the relevant conditioning states, the community converges on near-universal non-licensing, $C_t(u, c) \approx 0$. This is the formal counterpart of what earlier drafts (and LingbuzzPreprint.tex) call “near-zero entrenchment”.

3.5 INTERLOCUTOR MISALIGNMENT ABOUT CONDITIONING

Because c is construed, interlocutors can disagree about which conditioning state is in force. Let a hearer h maintain a posterior $q_h(c | e)$ over conditioning states given cues e (situational cues, ascription cues, stance cues, etc.). Then the hearer’s expected stability score is

$$\tilde{G}_t^{(h)}(u | e) = \sum_{c \in \mathcal{C}} \tilde{G}_t(u, c) q_h(c | e).$$

3.6 IDENTIFYING THE CONDITIONING STATE BEFORE PREDICTION

The conditioning state c is not a free parameter to be selected after a judgment is observed. For empirical work, c has to be fixed before predicting ratings or repair behaviour. The posterior $q_h(c | e)$ is the identification device: it is estimated from non-judgment cues such as activity type, medium, participant roles, speaker ascription, register cues, and overt norm orientation. A form like *might could* is therefore not rescued by choosing a sympathetic norm-centre after the fact; the model predicts different judgments only when the independent cues make that norm-centre probable.

This also blocks a competence/performance redux. C_t is identified by corpus-rate-per-opportunity, elicited production, and repair behaviour, not by acceptability ratings alone. F is identified by ratings, online measures, and self-reports. The model is falsified where these dissociate in the wrong direction: if satiation changes later corpus rates for a purportedly stable gap, if repair behaviour tracks prescriptive dissonance rather than licensing, or if high-opportunity preempted forms behave like low-evidence uncertainties, the state assignment is wrong.

Reassignment isn’t free, though. Localizing a failed prediction to a narrower conditioning state earns its keep only when that state is specifiable independently of the surviving cases – by activity type, medium, participant roles, or norm-orientation fixed before the failures were seen (Reynolds, 2026a). A partition of $\mathcal{C}$ redrawn around exactly the successes redescribes failure as success; it doesn’t rescue the model. Each study should therefore pre-register its conditioning partition and its estimation protocol for ρ_t^* (e.g. forced-choice norming over the competitor set, fit before the licensing data

is touched). Under that discipline, repeated post-hoc reassignment across studies counts against the framework itself, not merely against particular state assignments.

3.7 THE FEELING OF (UN)GRAMMATICALITY

Let $F_{i,t}(u, c) \in (-1, 0]$ be the subjective anomaly signal for individual i . Model it as a bounded response to an unbounded evidence total. The raw signal accumulates low stability, implementation costs, and ideological overlays:

$$A_{i,t}(u, c) = \alpha(1 - \tilde{G}_t^{(h)}(u \mid e)) + \gamma^\top R_i(u, c) + \delta D_i(u, c) + \varepsilon_i, \quad F_{i,t}(u, c) = -\frac{\max\{A_{i,t}(u, c), 0\}}{1 + \max\{A_{i,t}(u, c), 0\}},$$

where $h = i$ is the hearer processing the signal, and $\tilde{G}_t^{(h)}$ is the hearer's expected stability score accounting for situational uncertainty (§3.5). The bounded link keeps F in $(-1, 0]$ however large the summed costs grow, and the floor at zero encodes the absence of any positive feeling of grammaticality (§2.4.1). $R_i(u, c)$ is a vector of processing/repair costs (locality, interference, garden-path reanalysis, surprisal, etc.). The locality component $L(u)$ sums over the dependencies $d \in \mathcal{D}(u)$:

$$L(u) = \sum_{d \in \mathcal{D}(u)} \ell(|d|), \quad \ell(k) = \begin{cases} k, & k \leq K_{\text{sat}}, \\ K_{\text{sat}} + \beta(k - K_{\text{sat}})^\eta, & k > K_{\text{sat}}. \end{cases}$$

Here $|d|$ is the length of dependency d (in intervening discourse referents or words), K_{sat} is a saturation threshold beyond which additional distance contributes sublinearly, and β, η (with $0 < \eta < 1$) control the rate of that sublinear growth. $D_i(u, c)$ is prescriptive dissonance.

INTERPRETATION. $F = 0$ corresponds to a null affect (no anomaly reported), while $F < 0$ is a metacognitive warning proportional to the summed evidence of misfiring. The same feeling can arise from different mixtures of causes (structural, procedural, or social).

MEASUREMENT STRATEGY. F can be measured via explicit ratings (mapped monotonically to $(-1, 0]$) or implicit probes like N400/P600 magnitudes and self-paced reading slowdowns (proxies for components of R_i).

3.8 MEASUREMENT MODEL FOR C_t

$C_t(u, c)$ is latent and is estimated from converging indicators rather than from ratings. Let $y_t(u, c)$ be a vector of observable traces (corpus rate per opportunity, elicited production probability, repair probability, and recognition as a legitimate resource). Repair probability enters with the sign predicted by §3.3.2: low- C_t resources should invite repair when participants orient to the relevant norm-centre. A minimal measurement model is

$$y_t(u, c) = \mathbf{b} + \mathbf{\Lambda} \logit(C_t(u, c)) + \epsilon.$$

The corpus-rate-per-opportunity indicator is not a thought experiment. Figure 4 shows one worked case: agreement on collective-partitive subjects (*a bunch of people, the majority*), where surface-head number (singular) and notional number (plural) diverge. Counting only subject-position tokens after KWIC filtering (so non-subject false positives are removed) gives a plural-agreement share with a Wilson interval for each cell. The shares sit far above parity in every cell, with the interval width tracking the token count rather than the direction. This is what a licensing estimate for a contested operator contrast looks like when it is read off usage per opportunity instead of off ratings: a latent C_t recovered from a converging indicator, with its uncertainty made explicit.

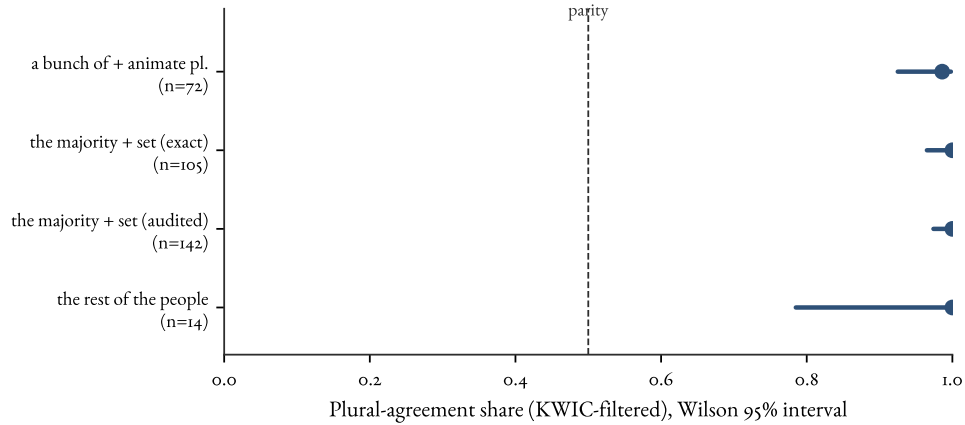


Figure 4: Corpus-rate-per-opportunity as an estimator of C_t for a contested agreement contrast. Plural-agreement share (dot) with Wilson 95% interval (bar) for collective-partitive subjects in COCA, after KWIC filtering to subject-position tokens; the dashed line marks parity (no directional preference). Data: agr-coca-projection probe (*bunch*, majority/minority, and partitive cells). Notional plural dominates in every cell, and the interval widens with the smaller denominators rather than drifting toward parity.

3.9 WORKED EXAMPLES: DERIVING STATUS FROM COMPONENTS

The status of the four canonical examples (§2.2) can now be defined by computing their component values.

1. **Can the have running* (Nonsense): $\text{map} = 0$ because structural analysis fails to assign a valid category structure. Result: $\tilde{G} = 0$. Status: Ungrammatical.
2. **I've finished it yesterday* (Semantic Clash): $\text{map} = 1$ (well-formed syntax), $C_t(g, c) \approx 1$ at the level of the component constructions (perfect, past adverbial), but $K \approx 0$ due to the clash between Present Perfect and specific past adverbial (temporal alignment ϕ_{temp} fails). Result: $\tilde{G} \approx 0$. Status: Ungrammatical.
3. *?friend of whose* (Situationally-Novel): $\text{map} = 1$, $K \approx 1$ (perfectly interpretable), but the licensing posterior stays diffuse: the opportunity set N_t is tiny, so preemption mass p_t barely accumulates and the posterior remains near its prior with wide dispersion – “not established” rather than rejected. Result: $\tilde{G}$ intermediate, low-confidence. Status: Marginal/Undefined.
4. **Which did you buy car?* (Stable Gap): $\text{map} = 1$, $K \approx 1$, but the licensing posterior is driven toward zero and concentrates there. Since N_t is large (questions are frequent) and the counterfactual choice probability ρ_t^* is high, the effective preemption mass p_t is large, collapsing both the posterior mean and its dispersion (Figure 5). Result: $\tilde{G} \approx 0$, high-confidence. Status: Categorically Ungrammatical.

4 DYNAMICS: WHY THE STATE TENDS TO BECOME WHAT IT IS

The dynamics module explains trajectories of $C_t(u, c)$ (and, in principle, refinements of the conditioning partition itself). It doesn't define grammaticality.

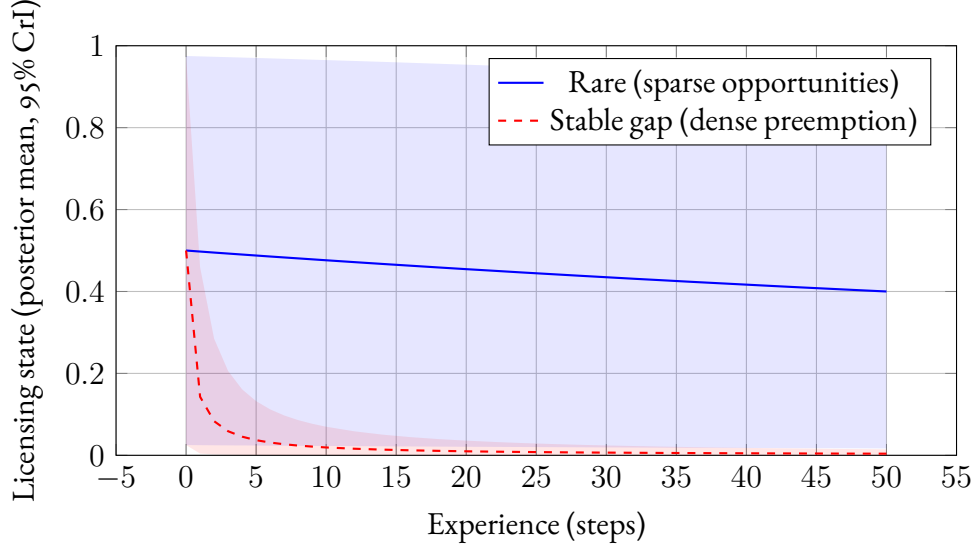


Figure 5: Licensing posteriors under preemption, from a $\text{Beta}(1, 1)$ prior with per-step effective preemption mass 0.01 (rare, blue) versus 5 (stable gap, red); lines are posterior means, shaded regions 95% credible intervals. The contrast is one of kind, not merely rate: the rare construction stays near its prior with a credible interval spanning nearly the whole unit interval – indeterminacy – while the stable gap’s posterior both falls and concentrates, collapsing onto confident rejection. The projections in §3.3.2 that turn on evidence structure (satiation, framing sensitivity, actuation) run over this dispersion difference, which the means alone don’t encode.

4.1 NICHES, COMPETITORS, AND OPPORTUNITY SETS

Let n index a CONSTRUCTIONAL NICHE (a communicative job), and let $\mathcal{V}_{n,t}^*$ be the opportunity set: the observed, licensed, and analogically generated variants that could plausibly do that job in some conditioning states. Let $N_t(n, c)$ be the number of opportunities for niche n in conditioning state c over some time window, and $k_t(v, n, c)$ the observed count of variant $v \in \mathcal{V}_{n,t}^*$.

The star matters. Stable gaps and innovations often involve variants with no positive tokens. Their counterfactual utilities are still defined because the hierarchy $\mathcal{H}$ generates analogical candidates. English learners can represent a left-branch extraction candidate by combining wh-fronting, modifier–noun packaging, and question formation, even if no one licenses that candidate as an English resource. The same machinery lets a novel but ordinary sentence inherit coverage from licensed higher-grain constructions rather than being preempted merely because that exact sentence type is new.

4.2 USAGE AS LICENSING $\times$ CHOICE AMONG CANDIDATES

Separate LICENSING from SELECTION. A variant can be licensed but rarely chosen.

Let $\rho_t^*(v \mid n, c) \in [0, 1]$ be the counterfactual probability of choosing v if it were available in niche n under c . A flexible choice model is a softmax over utilities:

$$\rho_t^*(v \mid n, c) = \frac{e^{U_t(v; n, c)}}{\sum_{v' \in \mathcal{V}_{n,t}^*} e^{U_t(v'; n, c)}}, \quad U_t(v; n, c) = \boldsymbol{\theta}^\top \mathbf{f}(v; n, c).$$

At the population level, the expected usage rate $\pi_t(v \mid n, c) \approx C_t(v, c) \cdot \rho_t^*(v \mid n, c)$. The counterfactual choice term can be high even when $C_t(v, c)$ is near zero; that is exactly the condition under which absence becomes informative.

4.3 DYNAMICS: PREEMPTION AS EFFECTIVE PREEMPTION MASS

The core idea is that learners update licensing not only from positive uses of u but also from structured non-occurrence when u is a plausible competitor and is repeatedly not chosen. The update shouldn't treat all "opportunities" as equally informative: absence is evidential only to the extent that u would have had non-trivial probability of being chosen if it were licensed.

For a target variant u associated with niche $n(u)$, define the EFFECTIVE PREEMPTION MASS in a time window as the expected number of u -tokens that would have been produced if u were a licensed competitor:

$$p_t(u, c) = \sum_{j=1}^{N_t(n(u), c)} \rho_t^*(u \mid n(u), c),$$

where $N_t(n(u), c)$ is the number of niche opportunities observed in that window. This quantity is large only when (i) the opportunity set is large and (ii) u would have been chosen at non-trivial rates if licensed.

A simple Bayesian learner for licensing represents each construction node (g, c) with a Beta posterior $\text{Beta}(a_{g,t}, b_{g,t})$, updated by three evidence streams: s_t (positive evidence), e_t (error evidence), and p_t (effective preemption mass). An observed token or missed opportunity distributes its evidence over activated nodes by the coverage weights $\lambda_g(u, c)$:

$$\begin{aligned} a_{g,t+1} &= a_{g,t} + \lambda_g(u, c) s_t(u, c), \\ b_{g,t+1} &= b_{g,t} + \lambda_g(u, c) \{e_t(u, c) + p_t(u, c)\}. \end{aligned}$$

The implied node-level mean licensing rate is then:

$$C_{t+1}(g, c) = \frac{a_{g,t+1}}{a_{g,t+1} + b_{g,t+1}}. \quad (29)$$

Token-level $C_t(u, c)$ is obtained by the partial-pooling equation in §3.2.

The streams themselves are ascription-weighted. The definition of Λ already excludes performance slips and nonce errors, and the conditioning state already fixes whose norms are being estimated; making this explicit, each token's contribution to s_t , e_t , or p_t is scaled by the hearer's ascription weight: the probability that the token is a competent realization of the norms of c , given cues about the producer (age, learner status, group membership) and about the token (disfluency, self-repair). Child and learner errors barely move adult posteriors even in error-dense environments, not because they go unheard but because they're attributed to development or transfer rather than to the norm-centre. The same machinery as $q_h(c \mid e)$ does the work: a token is evidence about the norms of whatever conditioning state its producer is taken to competently instantiate. Contact-driven change (koinéization) then falls out as a shift in ascription weights rather than a separate mechanism.

The two kinds of evidence also attach at different grains. Positive evidence distributes over activated nodes by the coverage weights: a heard token confirms every construction that covers it. Error

and preemption mass instead concentrate on the most specific node that individuates the candidate within its niche, because that node is what distinctively predicts the missing tokens. In Bayesian terms, the likelihood penalty for structured non-occurrence falls on the hypothesis that wastes probability mass on sentence types never observed; likelihood is a quantitative measure of the weight of implicit negative evidence (Perfors et al., 2011). So wh-fronting, question formation, and modifier-noun packaging keep their high licensing – each is confirmed by its own abundant tokens – while the left-branch-extraction configuration node, the one node that distinctively predicts **Which did you buy car?* tokens, absorbs the preemption mass from every fronting opportunity where a contiguity-preserving competitor wins.

This asymmetry also answers an acquisition worry. If token-level licensing were just pooled parent licensing, learners should pass through a stage of treating left-branch extraction as live. They don't need to: the candidate is generated by analogy, so it's on the table with non-zero prior support, but its dedicated node has no scaffolding of its own, fronting opportunities are dense in the input, and each one adds preemption mass. A companion corpus study quantifies the evidential situation: across 1,228 fronting opportunities in a sixteen-corpus Universal Dependencies sample, zero genuine determiner-head discontinuities occur (95% upper bound 0.24%), while contiguity-preserving alternatives – pied-piping, fused-head construals, and the *big mess* family – occur at measurable rates (Reynolds, 2026d). Comprehension pushes the same way: a fronted *whose* or *which* defaults to a fused-head parse, so the discontinuous analysis also loses the parsing race (Culicover et al., 2022). The non-satiation of left-branch extraction, confirmed in a meta-analysis of 25 experiments (Lu et al., 2024), is exactly what a concentrated near-zero posterior predicts (§3.3.2).

From decaying-gain learning to logistic dynamics. The discrete update (29) is the primary dynamics. Applied to a fixed environment (exogenous evidence streams), it is a decaying-gain estimator: the posterior mean follows a hyperbolic trajectory of the form $a_0/(a_0 + b_0 + \bar{e}t)$, as plotted in Figure 5, and converges without bifurcating. The logistic approximation $\dot{C} = r(g, c)C(1 - C)$ arises when the loop in §3.3.2 is closed: positive evidence is generated by production at a rate that grows with the current licensing state, while error and preemption mass accrue when the target isn't chosen. Replacing those endogenous evidence streams by their expectations yields a logistic drift, with $r(g, c)$ a net evidence rate and a gain that is constant under bounded evidence memory (decaying otherwise). I use this ODE only for qualitative fixed-point and comparative-statics arguments; the fixed points at 0 and 1 belong to the closed-loop population dynamics, not to individual open-loop learning.

4.4 EMERGENT CATEGORICALITY AS BIMODALITY

The sense of EMERGENCE used here is weak and mechanistic. Population-level categoricity is derived from learner-level updating, selection, and repair; it is not a label for whatever remains unexplained. This differs from Hopper's "emergent grammar", where grammar is perpetually provisional (Hopper, 1987). The present model allows conditioned states with near-absorbing attractors.

In the mean-field approximation, $\dot{C} = rC(1 - C)$ has fixed points at 0 and 1. When $r > 0$, positive evidence dominates and the licensing posterior is driven toward 1. When $r < 0$, error evidence and effective preemption mass dominate and the posterior is driven toward 0. Across construction nodes, a community with many evidence rates bounded away from zero therefore develops a

bimodal distribution of C_t : most nodes cluster near licensed or non-licensed attractors, while only low-evidence or contested nodes remain diffuse.

This is the formal sense in which categorical grammaticality emerges. Closure raises ρ_t^* by keeping the competitor set small and role-specific; large $N_t(n, c)$ supplies repeated opportunities; their product is the effective preemption mass. Closed, high-opportunity, update-critical contrasts are therefore exactly the parameter regime in which the licensing posterior becomes bimodal. A finite-population stochastic version should strengthen this pattern: sampling drift near 0 and 1 makes the boundaries sticky, while middle states remain vulnerable to renewed evidence. This connects the present dynamics to complex-adaptive and cultural-evolutionary accounts of language change (Beckner et al., 2009; Blythe & Croft, 2012; Kirby et al., 2014), and to invisible-hand explanations, where macro-level order emerges as the unintended consequence of micro-level choices (Keller, 1994).

4.5 HOW APPARENT CATEGORICAL GAPS ARISE WITHOUT HARD BANS

When $\text{map}(u, c) = 1$ and $K(u, c)$ is high but $C_t(u, c)$ is driven toward zero by persistent preemption in a large opportunity set, speakers converge on a sharp, non-satiating rejection profile. Additional processing penalties in $R_i(u, c)$ (e.g.

systematic garden-path repair due to an entrenched fused-head construal) can strengthen the subjective categoricity without any independent structural veto being posited.

This approach transforms the framework from a descriptive theory into a quantitative model capable of precise predictions about grammaticality judgments and their evolution over time.

5 THEORETICAL IMPLICATIONS

OVMG yields several important implications for the understanding of grammaticality:

First, it supports treating grammaticality as an emergent property in the formal sense of §4.4: discrete-looking judgments arise when licensing posteriors over constructional nodes become bimodal. Different patterns in the distribution of grammatical constructions across adjective types illustrate the same point. The distribution of *much* versus *more* with different adjective classes (comparative governors, participial adjectives, etc.) follows real patterns, but those patterns arise from interactions among modifier semantics, adjective scale structure, and discourse-pragmatic factors (Reynolds, 2024b). This helps explain why efforts to reduce grammaticality to simple necessary and sufficient conditions have repeatedly fallen short.

Second, this framework clarifies how formal and usage-based approaches capture different aspects of grammatical stability. The stability of grammatical patterns depends both on their internal systematic properties (emphasized by formal approaches) and on their role in meeting communicative needs (emphasized by functional accounts). Many grammatical phenomena exhibit stability patterns that can't be reduced to local collections of features. Discourse context can change whether a construction maintains stable form-value relationships, because the relevant operator values are conditioned by the activity, norm-centre, and public-update role. This helps explain why purely local syntactic models often fail to capture the full range of grammaticality phenomena.

Third, the analysis suggests specific predictions about how grammatical stability is maintained and lost. If grammatical constructions are maintained through multiple interacting factors, one should expect:

1. Instability to manifest in coordinated ways across multiple properties rather than through isolated changes

2. Periods of gradually increasing instability followed by relatively rapid reorganization when stability thresholds are crossed
3. Different but equally stable grammatical patterns arising in different language communities
4. Gradient effects in grammaticality judgments reflecting varying degrees of stability

Fourth, OVMG illuminates the relationship between competence and performance. Rather than treating these as fundamentally different phenomena, they can be understood as different manifestations of the same stability conditions. Processing limitations and other performance factors help shape which form–value relations become stable, while those stable relationships in turn constrain possible performance patterns. The identification strategy in §3.6 keeps this from becoming an escape hatch: licensing and feeling are measured by different evidence streams and can falsify each other.

Finally, this approach offers a new perspective on systematic constraints in grammar, such as the English ban on left-branch extraction. Instead of viewing these as either innate rules or processing limitations, they can be understood as particularly robust stability conditions in form–value relations. Their persistence reflects deep patterns of stability, while cross-linguistic variation shows how different stable solutions can emerge in different communities.

These implications suggest concrete directions for future research. More detailed studies are needed of how multiple properties interact to create and maintain stable form–value relations, better methods for measuring degrees of stability, and closer examination of the transition points where grammatical systems reorganize. The framework also calls for renewed attention to variation across language communities, as different stability patterns may shed light on the fundamental nature of grammatical organization.

The central idea is that grammaticality supports a kind claim in the projectibility-first sense: a category earns standing by tracking a profile whose partial observation licenses declared inferences, with stabilizing support graded by the evidence (Boyd, 1999; Reynolds, 2026a). Here the profile is conditioned form–value stability; the projections are repair rates, transmission fidelity, satiation profiles, and change trajectories (§3.3.2); and the support grades are explicit. Bimodal licensing dynamics supply stabilization, the production–repair loop is a maintenance claim, and corrective control – the strict reading of HOMEOSTATIC – remains an open empirical question rather than a label inherited from the framework (Reynolds, 2026b). The kind claim earns its keep by those projections, not merely by naming a cluster of mechanisms.

5.1 ACTUATION AND THE DYNAMICS OF GRAMMATICAL CHANGE

The concept of *actuation* can now be defined as a sign change in the expected evidence balance for licensing. Informally, actuation occurs when the expected positive stream $s_t(u, c)$ begins to outweigh the combined negative streams $e_t(u, c) + p_t(u, c)$ across successive windows, so that the posterior mean $C_{t+1}(u, c)$ in (29) increases rather than decays. In the mean-field approximation (box in §4.3), this corresponds to the regime in which the net evidence rate $r(u, c)$ becomes positive, yielding the familiar S-curve at the population level.

The factors that drive such bifurcations align with the motivations discussed in §2.5. Semantic reanalysis may increase utility (and thus ρ_t^*) by making a form–value relation more transparent or useful. Social pressures may shift utilities (prestige effects) or alter the relevant community standard. Structural motivations such as analogical extension can systematically raise licensing probability by borrowing strength from frequent neighbors. Processing innovations may reduce error rates (e_t) or locality costs, improving the net evidence balance.

Actuation isn't simply a matter of individual innovation but requires coordinated community-level change. A few speakers adopting a marginal construction doesn't guarantee its success; actuation demands that the underlying dynamics shift such that acceptance systematically outweighs rejection across the speech community. This explains why many innovations fail despite being individually sensible – they appear before the community conditions are right for the evidence balance to become positive.

The bifurcation framework also clarifies why certain changes appear to accelerate once they begin. Near the critical point where evidence streams balance, small perturbations in community attitudes can trigger rapid shifts between rejection and acceptance. As the evidence balance becomes more strongly positive, the construction moves further from the unstable rejection state, making reversal increasingly unlikely. This creates the characteristic S-curve pattern observed in many documented language changes (Kroch, 1989), derived from the population dynamics in §4.4.

For constructions currently existing in the marginal zone – those with near-zero evidence balance – the framework predicts heightened sensitivity to external factors and greater cross-community variation. Small speech communities may show particularly volatile behaviour near bifurcation points, as stochastic effects can overwhelm weak deterministic trends. This may explain why certain changes appear to originate in geographically or socially peripheral communities before spreading to larger population centres.

5.2 MACRO-TYPOLOGICAL CONSTRAINTS ON GRAMMATICAL DESIGN

Recent macro-typological work sharpens the question of how strongly grammar is constrained across languages. Verkerk et al. (2025) test 191 implicational universals from the Universals Archive against Grambank's 2,430-language morphosyntactic sample, using Bayesian models that control explicitly for genealogical and areal non-independence and then follow up with spatiophylogenetic analyses of evolutionary rates. A naïve analysis that ignores relatedness appears to support the vast majority of proposed universals, but once phylogeny and geography are accounted for, only about a third (60 of 191) remain statistically supported. Support is concentrated in relatively narrow domains: most hierarchical universals and a substantial minority of “narrow” word-order universals survive, whereas “broad” word-order universals and miscellaneous others fare poorly. Diachronically, supported universals typically correspond to “harmonic” combinations of features (for instance, consistent head-dependent orders) that languages are more likely to evolve into than out of, with these preferred configurations recurring independently across lineages.

From an OVMG perspective, these findings identify recurrent attractors in the global design space of operator-valued form–value relations rather than exceptionless grammatical laws. On the present account, such attractors correspond to configurations for which the net evidence balance (positive evidence s_t vs. preemption/error mass $e_t + p_t$) in the dynamics of §4.3 is positive across a wide range of communities. They are cognitively and communicatively favourable enough that repeated episodes of change tend to push $C_t(u, c)$ toward fixation in lineage after lineage. The fact that roughly two-thirds of the tested universals fail once autocorrelation is controlled for also fits a de-idealized view of grammaticality. Community-specific conventions still have considerable freedom in how they realize operator values, with only some regions of the space strongly preferred. Large-scale comparative work of this kind therefore complements OVMG: it circumscribes which parts of the operator-value design space are globally stable, while the present framework provides a meso-level account of how local community dynamics and form–value coherence give rise to those long-run

macro-typological regularities.

5.3 PREDICTIONS

The model's distinctive predictions come from separating positive evidence, opportunity structure, and counterfactual choice probability. Cross-linguistic gender remains a useful boundary case, but it is not the strongest test. Any framework that treats grammaticalized concord as obligatory predicts stronger judgments where concord is more tightly grammaticalized. The companion operator-stratum account also complicates the simple gender story: gender concord can be policed through paradigm inertia even when its public-update value is attenuated (Reynolds, 2026c). The sharper predictions concern stable gaps, artificial learning, and L2 transfer.

SATIATION FRAMING. Repeated exposure should attenuate negative feelings most strongly when a form is low-evidence rather than strongly preempted. Independent relative *whose*, as in **The packages are still here, but Nathan, whose was open, just left.*, should be sensitive to framing: participants told that tokens come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told that the same tokens are errors. Left-branch extraction should resist the same treatment because its opportunity set is large and its effective preemption mass is high. The dependent variable is not only rating change but later repair behaviour and elicited production.

ARTIFICIAL-LANGUAGE LEARNING. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target variant is never produced. In the high-opportunity condition, competitor variants repeatedly occupy the relevant niche and the target has high ρ_i^* ; in the low-opportunity condition, the niche rarely arises or the target would have low counterfactual utility. OVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. This puts numbers on the Figure 5 contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 PREEMPTION TRANSFER. Early L2 rejections should track L1 preemption mass, not only L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, $q_h(c | e)$ and the constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker's (1972) concept of INTERLANGUAGE, but gives a more specific prediction: transfer is strongest where the L1 has accumulated large effective preemption mass.

INTONATIONAL OPERATORS. Nothing in the model privileges segmental material, so tunes are predicted operator exponents wherever they form closed, high-opportunity, update-critical repertoires: the English polar-question rise and nuclear-accent placement configure what an utterance does to the common ground much as tense and agreement do (Reynolds, 2026c). The test is whether tune-text misassignment (e.g. nuclear accent on a function word under broad focus) is policed like grammar – categorical rejection, repair, resistance to framing – or like style. A worked analysis parallel to the Turkish appendix is future work; the prediction stands either way, and failure would count against the channel-agnosticism that motivates the model's name.

5.4 RELATIONSHIP TO GENERATIVE GRAMMAR

The generative tradition has provided linguistics with foundational contributions into the systematic nature of grammatical knowledge. Most fundamentally, it demonstrates that grammaticality can't be reduced to semantic plausibility, processing ease, or frequency of attestation. When Chomsky introduced *Colorless green ideas sleep furiously*, he showed definitively that speakers can recognize syntactically well-formed sentences even when they are semantically bizarre. The framework coherently explains categorical constraints like the impossibility of extracting determiner-adjective sequences in English (**Which do you prefer car?*), and critically, it captures the fact that such sequences remain ungrammatical even when their intended meaning is clear and processing demands are low.

OVMG shares with generative grammar several foundational observations: that grammaticality judgments reflect systematic, real patterns (Dennett, 1991); that these patterns can't be reduced to meaning or processing alone; and that certain syntactic configurations appear to be categorically excluded regardless of context. The analysis of multiple center embeddings presented in §2.3.1 builds directly on ideas about recursive structure, and my treatment of stable gaps (§2.3.1) acknowledges the generative discovery that some constructions appear to be universally excluded by the grammar itself.

The generative tradition has also identified fascinating puzzles that any theory of grammar needs to address, such as the independent relative *whose* (1d). As Hankamer and Postal (1973) observe, this construction appears to violate no syntactic principles: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and *whose* functions perfectly well as a dependent relative pronoun (*the student whose friend was visiting*). The generative tradition's careful documentation of such cases, where seemingly parallel constructions show puzzlingly different grammatical status, has been invaluable in pushing theoretical development forward.

Where OVMG departs from generative grammar is in its explanation of such patterns. Rather than positing an autonomous syntactic component, OVMG suggests that grammatical constraints are produced by the interaction of form–value relations within specific communicative situations. The independent relative *whose* construction illustrates this difference. For this construction to be felicitous, multiple conditions have to converge: the possessor has to be sufficiently accessible in the discourse while the possessum is predictable enough to license ellipsis, but the possessive relationship needs to be semantically significant enough to warrant explicit marking, and this configuration has to occur in a context where a relative clause is the optimal way to package this information. The extreme rarity of contexts satisfying all these conditions appears to prevent the construction from becoming conventionalized in the grammar at all – speakers encounter it so rarely, despite perfectly common components, that even when all conditions align perfectly, the construction feels alien.

This approach draws on the generative observations about systematic constraints while providing a different perspective on their source. Where generative theory has to explain why a syntactically possible and pragmatically useful construction is systematically avoided, OVMG suggests that extreme mismatches between predicted and observed frequency may themselves be evidence of strong negative weighting, even when the exact source of that evidence remains unclear. The framework thus preserves what is most valuable in generative theory – its recognition of systematic grammatical constraints – while embedding those insights in a broader theory of how form–value relations become established and maintained in language communities.

Other cases that generative grammar struggles to explain are those that are grammatical with one

meaning but ungrammatical with another, such as (13) *I have 16 years*. While syntactically identical to grammatical expressions like *I have 16 dollars*, this construction becomes ungrammatical specifically when used to express age. A purely syntactic account would have to explain why the same structure is well-formed in one case but ill-formed in another, despite no apparent syntactic differences. OVMG, in contrast, locates the source of ungrammaticality in the community's form–value conventions: *have*+numeral years has become conventionalized for expressing duration or future time (*I have 16 years until retirement*) but lacks positive evidence for expressing age, where a different construction (*I am 16 years old*) is the established pattern. Similar cases arise with plural forms that are grammatical with some meanings but not others (e.g. *peoples* for ethnic groups but not multiple individuals) and with verbs that resist certain arguments despite no obvious syntactic prohibition (e.g. *discuss about*). These meaning-dependent grammaticality patterns suggest that what gets down-weighted or up-weighted often depends on specific form–value associations rather than purely structural constraints.

5.5 RELATIONSHIP TO CONSTRUCTION GRAMMAR

Construction Grammar (CxG) (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012) represents a significant theoretical advance in the understanding of linguistic knowledge. At its core, CxG argues that language consists of learned relationships between form and meaning at multiple levels of complexity. These form–value relations, or constructions, range from individual morphemes to abstract syntactic patterns. This perspective helps explain phenomena that proved challenging for earlier approaches, including idiomatic and partially schematic expressions that don't fit a clean core/periphery split.

CxG's most valuable contribution lies in demonstrating that meaning suffuses all levels of grammatical organization. Rather than treating syntax as an autonomous formal system that interfaces with semantics only at designated points, CxG reveals how meaning and form are inseparable aspects of linguistic knowledge. For instance, the *What's X doing Y?* construction carries an implication of incongruity that can't be derived from its component parts (Kay & Fillmore, 1999). Recent work connects this framework to model-internal analysis: Weissweiler et al. (2023) use construction grammar to probe how neural language models handle different levels of linguistic abstraction.

OVMG shares these CxG commitments about form–value relations and constructional abstraction. Its difference is not that one substance class is intrinsically privileged. The companion operator-stratum paper explicitly denies a substance-based boundary: phonological, gestural, lexical, morphological, and syntactic material can all realize operator contrasts when they enter a closed, accountable public-update repertoire (Reynolds, 2026c). The OVMG claim is narrower. Categorical grammaticality clusters around form–value relations with a particular role: they are high-opportunity, closed-paradigm contrasts that configure update, role allocation, scope, reference, or repair.

This difference becomes clear when speakers judge violations. Register, politeness, genre, and many lexical choices can be stable and socially important without blocking the public update itself. A wrong tense, agreement value, clause-linking marker, or obligatory allomorph can mis-set the update instruction even when the intended message is recoverable. The age-predication example (*I have 25 years*) is transparent but not licensed for that English frame; its rejection comes from non-licensing in a high-opportunity constructional niche, not from the substance of syntax or morphology as such.

OVMG therefore preserves CxG's systematic treatment of constructions while adding a prediction about which constructional contrasts attract categorical policing. The relevant contrasts are

those whose closure and opportunity structure drive the licensing posterior toward the bimodal regimes in §4.4. That lets CxG’s continuum of constructions coexist with a non-arbitrary explanation of why some violations are heard as grammar and others as style, stance, or appropriateness.

5.6 RELATIONSHIP TO USAGE-BASED APPROACHES

OVMG shares with usage-based approaches (UBA) (J. Bybee, 2006, 2010; J. L. Bybee, 2007) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context.

Two distinctions that organize the formal core have clear precedents in this tradition, and it is worth marking the debts precisely rather than claiming novelty for the distinctions themselves. The individual/population split – $\Lambda_{i,t}$ for a speaker’s own licensing state, C_t for the population rate – restates Schmid’s (2020) separation of ENTRENCHMENT (the cognitive reorganization of a speaker’s knowledge) from CONVENTIONALIZATION (the social establishment of a community’s regularities), the two coupled feedback cycles of his Entrenchment-and-Conventionalization model. The population dynamics themselves descend from Croft’s (2000) utterance-selection theory, in which variants are replicated and selected within a community; the S-curve results the model reuses are already worked out in that lineage (Blythe & Croft, 2012). Frequency per opportunity is likewise not new: it is the core of Stefanowitsch’s (2006) case that corpora contain negative evidence, where a form counts as “significantly absent” rather than “accidentally absent” only once its observed frequency is weighed against the frequency expected given its opportunities. Variationist sociolinguistics has normalized by opportunity since Labov’s principle of accountability (Labov, 1972).

What OVMG adds is not these distinctions but their joint formalization and a prediction. The licensing-versus-selection factorization $\pi_t \approx C_t \cdot \rho_t^*$ turns Croft’s replication-versus-selection into an estimable product; effective preemption mass $p_t = N_t \cdot \rho_t^*$ puts a quantity on the gap that Stefanowitsch left open, since he showed how to establish that a form is significantly absent but stressed that significant absence “does not, in itself, provide any clues as to why”. Licensing-versus-selection is exactly that missing clue: a significantly absent form may be unlicensed ($C_t \approx 0$) or licensed but dispreferred ($\rho_t^* \approx 0$), and the two project differently. The Beta dynamics then convert these quantities into the bimodality result (§4.4) and the prediction about which contrasts attract categorical policing. The relation to Goldberg’s (2019) COVERAGE and statistical preemption is one of formalization in the same spirit: her “explain me this” puzzle – a high-coverage double-object frame preempted by the prepositional dative – is a preemption-mass story in which the competitor wins a high-opportunity niche. One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single construction-competition process. OVMG keeps them analytically separate – error evidence e_t and preemption mass p_t are distinct streams – but they enter the Beta update additively (§4.3), so the model can represent Ambridge and colleagues’ single-process view as their sum while leaving the separability an empirical question rather than a stipulation.

The analysis of independent relative *whose*, as in *‘I saw Joan, a friend of whose was visiting* is a case in point. A simple UBA account might predict that this construction’s marginality stems from its low frequency. But this explanation proves insufficient: the construction isn’t merely rare but dramatically rarer than one would expect given the frequency of its component parts. Independent

whose appears in interrogatives (*Whose is that?*), and the relative *whose* is common in dependent contexts (*the student whose paper was late*). Given these frequencies, analogical extension should make the independent relative use more common than it is. The extreme rarity of independent relative *whose* in corpora, despite the grammatical availability of comparable elements and contexts, marks it as more than just infrequent – it points to a systematic gap in form–value relations.

The difference is not that OVMG exempts the operator stratum from usage dynamics. It makes the dynamics more specific. A rare form can remain grammatical when the opportunity set is tiny or when higher-grain constructions license the token by partial pooling. A transparent candidate can become categorically rejected when the opportunity set is large, the analogical candidate would have had high ρ_t^* , and an entrenched competitor wins every time.

This position lets OVMG incorporate usage-based findings about frequency and entrenchment while predicting when frequency becomes preemption. Cases like independent relative *whose* demonstrate the need for a theory that distinguishes simple rarity from systematic non-licensing. The same machinery explains why high-opportunity operator contrasts often yield sharp judgments: the evidence stream is dense enough for effective preemption mass to accumulate.

5.7 RELATIONSHIP TO LOGICALITY OF LANGUAGE ACCOUNTS

An alternative perspective, prominent in recent formal semantics, suggests certain types of unacceptability stem from the language faculty itself possessing a deductive system that identifies and filters sentences with logically trivial meanings (tautologies or contradictions) (cf. Chierchia, 2013; Del Pinal, 2019; Fox, 2000). This “logicality of language” hypothesis attempts to explain, for example, systematic restrictions on quantifiers by arguing the unacceptable cases are “L-trivial” – their triviality arises solely from the meaning and configuration of logical/functional terms (like *every*, *some*, *not*), irrespective of the open-class words (like *student*, *run*) (Gajewski, 2009). A key challenge is explaining why simple tautologies or contradictions (e.g. *It is raining and it is not raining*) are often acceptable. Del Pinal (2019) argues against “Logical Skeletons” (which assume the system ignores open-class word identity) in favour of “LF+RESCALE”, a view where the system sees standard logical forms but allows optional, context-dependent modulation of open-class terms (e.g. interpreting the second ‘raining’ as ‘raining hard’) to yield non-trivial meanings.

OVMG offers a potentially broader and more unified account. While ‘logicality’ approaches excel at explaining restrictions tied to functional vocabulary via L-triviality, OVMG aims to cover a wider spectrum of ungrammaticality, including cases not easily reducible to logical contradiction, such as absent form–value relations (1a), strong deviations from conventional community forms (17), or extreme, unexpected rarity (1d). By grounding grammaticality in community-specific conventions (§2.1.1) and allowing for gradient compatibility ($K(u, c)$) and acceptance ($C_t(u, c)$), OVMG inherently accommodates cross-linguistic variation and degrees of acceptability, aspects less central to the L-triviality filter. While LF+RESCALE provides a specific mechanism for acceptable ‘trivialities’, OVMG suggests these might arise from more general principles of interpretation within community norms, potentially avoiding the need to posit a dedicated deductive module and specific operators like RESCALE, and offering a clearer distinction between objective grammatical status and subjective processing effects or ‘feelings’ (§3.7). Thus, OVMG frames grammaticality as an emergent consequence of communicative practice in the formal sense of §4.4, rather than a direct output of logical computation within syntax.

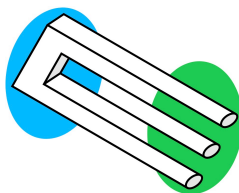


Figure 6: Impossible trident

5.8 RELATIONSHIP TO RELEVANCE-THEORETIC ACCOUNTS

Recent work by Scott-Phillips (2026) offers a fundamental observation: linguistic intuitions about acceptability arise as byproduct effects of cognitive systems for interpreting communicative acts. Just as people immediately sense when a visual stimulus violates core assumptions about physical objects (as with impossible objects), they detect when utterances violate basic presumptions about communicative efficiency. Significantly, Scott-Phillips argues that unacceptability occurs not from mere inefficiency, but from an inherent impossibility of interpreting an utterance consistently with these presumptions – similar to how an impossible trident (Figure 6) can't be interpreted as physically cohesive in any context.

OVMG shares several key premises with this account. Both frameworks reject the need for an innate grammar faculty, locating linguistic intuitions instead within general cognitive systems. Both recognize that language develops from communicative needs rather than autonomous syntactic principles. OVMG's emphasis on situation-specific form–value relations builds directly on Scott-Phillips's arguments about how communicative pressures shape linguistic conventions.

The frameworks differ primarily in their explanatory mechanisms. Where Scott-Phillips argues that grammaticality judgments reduce to impossibilities of efficient interpretation, OVMG suggests that while communicative pressures shape which form–value relations become conventionalized, these relationships then create systematic constraints that can't be reduced to efficiency alone. For Scott-Phillips, one would have to demonstrate that (1c) contains inherent contradictions making efficient interpretation impossible. OVMG instead analyzes how the present perfect's tense–aspect operator value clashes with the meaning of the lexeme *yesterday*. While these analyses might ultimately converge, it remains unclear why the criterion of inherent impossibility of interpretation should apply specifically to operator-value violations rather than to lexical-lexical conflicts or certain phonological patterns. The scope of what constitutes an interpretive impossibility requires further theoretical development.

This difference has important empirical implications. OVMG makes relatively concrete demands: it is possible to test whether specific form–value relations are stable within a communicative situation and identify precise points of operator-value conflict. The challenge for relevance-theoretic accounts, as Scott-Phillips (personal communication, Dec. 16, 2024) acknowledges, lies in establishing independent, empirically vulnerable claims about what makes efficient interpretation inherently impossible rather than merely difficult. Future work comparing specific predictions of each approach – particularly around how novel constructions become acceptable or unacceptable – could help clarify their relationship and complementary insights.

6 LIMITATIONS

While the proposed framework offers a comprehensive account of grammaticality, integrating insights from various theoretical traditions, it's not without limitations. One potential limitation lies in the framework's reliance on the concept of "communicative situation". Defining the relevant conditioning context and determining the precise set of shared norms that govern grammaticality judgments can be challenging. As noted in §2.1.1, speakers navigate overlapping contexts that shift on multiple dimensions – immediate situation, durable social positions, and the norms they orient to. The framework acknowledges this fluidity and heterogeneity, but further research is needed to develop more precise methods for operationalizing these dimensions and measuring their influence on grammatical stability.

The framework, in its current form, may not fully capture the complexities of stylistic variation and individual preferences. While it accounts for broad patterns of acceptability and rejection, it doesn't delve deeply into the nuances of stylistic choices that fall within the realm of grammatical acceptability. Future refinements could incorporate a more fine-grained model of stylistic variation and its interaction with core grammatical principles.

Another potential limitation concerns the framework's ability to account for purely formal constraints that seem to exist independently of meaning or communicative function. While the paper argues that many seemingly arbitrary restrictions can be explained by historical processes and the interplay of various motivations, there may be residual cases of purely formal constraints that resist explanation within the current framework. Further investigation into these cases could lead to a more complete understanding of the factors shaping grammatical systems.

Despite these limitations, the proposed framework provides a robust and flexible foundation for understanding the multifaceted nature of grammaticality. It offers a promising avenue for future research, and the limitations outlined here serve as points of departure for further refinement and elaboration.

7 METHODOLOGICAL CONSIDERATIONS FOR FUTURE RESEARCH

The development of experimental syntax since the 2000s (Schütze, 2016) has transformed how gradient acceptability is measured, letting researchers quantify subtle differences in judgments and track how they change with exposure. The framework's predictions invite corpus-based, experimental, and cross-linguistic investigation. Several avenues stand out:

1. *Corpus analysis*: Large, balanced corpora allow researchers to track frequency patterns and stability over time. Investigating rare or marginal forms (e.g. the independent relative *whose*) in corpora can reveal whether low frequency corresponds to genuinely unstable operator-value patterns or merely a sampling gap. Comparative corpus research in multiple languages can test predictions about how community values influence grammatical distinctions.
2. *Grammaticality judgment experiments*: Controlled psycholinguistic tasks, including magnitude estimation or forced-choice paradigms, can assess subtle gradience in grammaticality judgments. By manipulating exposure and increasing familiarity with marginal forms, researchers can measure satiation effects and distinguish between stable categorical constraints and constructions that become more acceptable through repeated exposure.
3. *Processing measures*: Eye-tracking, self-paced reading, and EEG studies can identify whether some judgments arise from processing overload rather than stable grammatical constraints. If a form

becomes easier to parse with practice, it suggests that processing complexity, rather than an immutable rule, lies behind initial judgments of ungrammaticality.

4. *Sociolinguistic fieldwork*: Investigations in communities with distinct dialects or multilingual practices can clarify how social and pragmatic motivations shape grammatical stability. Elicitation tasks and careful comparisons across speech communities can confirm that what counts as grammatical depends on local norms rather than universal principles.
5. *Longitudinal and historical studies*: Diachronic corpora and historical grammars can trace how marginal constructions evolve, testing predictions about which factors lead certain patterns to stabilize, which fade, and which undergo reanalysis. Tracking a community's acceptance of previously dubious constructions over time provides direct evidence for the gradual entrenchment processes posited by the framework.
6. *A threshold-reality test*: The categorical predicate G_t earns standing only if community treatment of a form changes *discontinuously* as the graded state crosses $\tau(c)$, over and above the smooth effect of $\tilde{G}_t$. That's a regression-discontinuity question. Hearer responses are naturally multinomial (accept, repair, request clarification, sanction), and covariate-adjusted local-likelihood estimators for discontinuities in multinomial response probabilities exist off the shelf (Lazzaro, 2026), with the processing-cost components of R_i entering as predetermined covariates. Two caveats are design constraints rather than afterthoughts: $\tau(c)$ is latent, so it has to be estimated, and pre-registered, before the outcome data is touched; and the bimodality result (§4.4) predicts that the density of forms near the threshold is thin, so power is scarce exactly where the test bites. The design also yields a further distinctive prediction: because $\tau(c)$ is decision-theoretic (§3.3), independently manipulating the stakes should move the location of the discontinuity. Smoothness through the threshold, by contrast, would demote G_t to bookkeeping over $\tilde{G}_t$.

Figure 7 states the test in one picture: the categorical predicate is real only if community treatment jumps at $\tau(c)$.

8 CONCLUSION

This paper proposes a novel framework for understanding grammaticality, one that moves beyond the traditional competence–performance dichotomy and embraces the dynamic interplay of form, meaning, processing constraints, and sociolinguistic factors. By conceptualizing grammaticality as an emergent property in the sense derived in §4.4, the framework accounts for both the categorical and gradient aspects of grammaticality judgments. It explains why some constructions are rigidly ungrammatical, others fluctuate between marginal and acceptable, and still others evolve into stable, conventionalized patterns.

The framework's core tenets – that grammaticality involves conventional form–value relations, that these relationships interact with processing constraints and sociolinguistic factors, and that different types of violations arise from different mismatches between form and meaning – provide a principled basis for understanding a wide range of linguistic phenomena. The analysis of examples drawn from both formal syntax and experimental data has demonstrated the framework's explanatory power, illuminating challenging cases that have resisted unified explanation in previous approaches.

The framework generates testable predictions about which ungrammatical constructions might change over time and offers potential applications for language teaching, clinical linguistics, and lan-

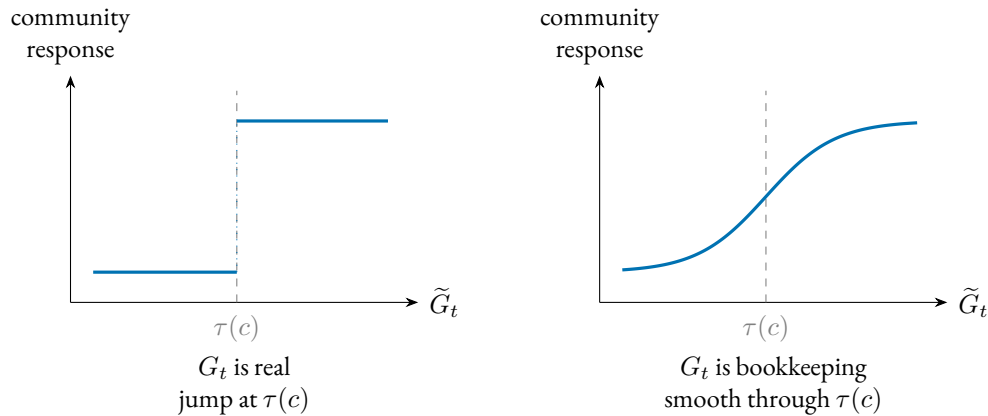


Figure 7: The threshold-reality test. If categorical grammatical status is a real cut in the population state, community treatment of a form should change discontinuously as $\tilde{G}_t$ crosses the situational threshold $\tau(c)$ (left); if the cut is only a summary of the graded score, treatment should vary smoothly through it (right). A covariate-adjusted regression-discontinuity design on multinomial hearer responses (Lazzaro, 2026) distinguishes the two.

guage documentation. It predicts that satiation effects should be more readily inducible for low-evidence constructions than for high-opportunity stable gaps, that artificial-language learners should distinguish zero evidence from strong preemption, and that early L2 rejections should track L1 preemption mass rather than L2 frequency alone.

Integrating the methodologies outlined in the previous section offers a path toward a more comprehensive understanding of how form–value relations become stable within communicative situations, what factors promote or inhibit their entrenchment, and how cross-linguistic variation arises. Future work can refine the metrics for identifying stable form–value mappings, develop computational models to predict the emergent regularities derived in §4.4, and expand the empirical base to underrepresented languages and language contact situations.

While acknowledging the limitations of the current formulation, this paper argues that the proposed framework represents a significant step toward a more comprehensive and nuanced understanding of grammaticality. By integrating insights from generative, functional, and usage-based approaches, it offers a unified perspective that recognizes the multifaceted nature of linguistic knowledge and its grounding in both individual cognition and usage within communicative situations.

Future research should focus on further refining the framework, developing more precise methods for measuring form–value stability, and investigating the complex interactions between different types of motivations (semantic, social, structural, and iconic). Cross-linguistic studies should prioritize contrasts whose operator status differs across languages, because the model predicts categoricity where closed, high-opportunity, update-critical repertoires form.

Ultimately, the study of grammaticality offers a window into the fundamental workings of human language. By de-idealizing grammaticality and embracing its dynamic, situation-relative nature, it becomes possible to gain a deeper understanding of the cognitive and social forces that shape language. This framework provides a robust foundation for such investigations, paving the way for a more integrated and comprehensive understanding of what it means for an utterance to be considered

part of a language.

A TURKISH VOWEL HARMONY AND OPERATOR-EXPONENT REALIZATION

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.³ Only the latter interacts with grammaticality in the sense of the OVMG because suffixal harmony is part of how an inflectional operator exponent is realized.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* ‘doctor’ violate backness harmony, but are fully acceptable; the language community simply memorizes the form, so $C_t(u, c) \approx 1$ and $K(u, c) \approx 1$.

- (30) doktor
 doctor
 ‘doctor’ (disharmonic stem, grammatical)

Because no operator feature is left unrealized, the form–value mapping succeeds and

$$\tilde{G}_t(u, c) = C_t(u, c) \cdot K(u, c) \cdot \text{map}(u, c) \approx 1.$$

SUFFIXAL HARMONY: OPERATOR-EXPONENT REQUIREMENT. Inflectional morphemes are lexicalized with an underspecified vowel; the correct allomorph has to copy [$\pm$ BACK] (and, for some suffixes, [$\pm$ ROUND]) from the final stem vowel. Using the “wrong” vowel leaves part of the exponent-selection requirement unsatisfied, so the null outcome in the K calculation wins and the word is ungrammatical.

- (31) a. kitap-lar
 book-PL.BACK
 ‘books’ (harmonic, grammatical)
 b. * kitap-ler
 book-PL.FRONT
 intended “books” (suffix harmony violation)
- (32) a. gül-dü
 laugh-PST.FRONT.ROUND
 ‘s/he laughed’ (harmonic, grammatical)
 b. * gül-du
 laugh-PST.BACK.ROUND
 intended “s/he laughed” (suffix harmony violation)

OVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

- *Mapping* succeeds: the plural / past node selects an exponent.
- *Compatibility* $K(u, c) \approx 0$: the chosen allomorph fails to realize the [BACK] feature copied from the stem.
- *Community acceptance* $C_t(g, c) \approx 1$ at the node level: the plural and past nodes are licensed for every speaker. What’s entrenched is the node, not the ill-formed token; node-level licensing pools to the token by the coverage equation in §3.2, and the failure lives in K .

³See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arık, 2015) for experimental evidence on native judgments.

Hence $G_t(u, c) = 0$ and speakers judge the word as categorically wrong, not merely odd-sounding. By contrast, *doktor* in (30) keeps $K(u, c) = 1$; harmony is a phonotactic preference that affects only the phonological well-formedness component of $F_{i,t}(u, c)$, not grammaticality.

LOCALIZED EXCEPTIONS. Some derivational suffixes (e.g. *-imsi* ‘-ish’) are lexically marked “disharmonic”. The community memorizes them with $C_t(u, c) \approx 1$, so no feature remains unrealized and $G_t(u, c) = 1$ despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

In sum, Turkish suffix harmony errors are grammaticality failures because they mis-realize a closed inflectional operator exponent. Stem disharmony merely lowers phonological well-formedness and so affects only the listener’s feeling $F_{i,t}(u, c)$.

If it could be shown that phonology alone caused feelings of ungrammaticality, without realizing an operator contrast or expression-shape requirement, that would constitute a serious challenge to this framework’s central claim about the functional target of grammaticality.

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REFERENCES

- Ambridge, B., Bidgood, A., Twomey, K. E., Pine, J. M., Rowland, C. F., & Freudenthal, D. (2015). Preemption versus entrenchment: Towards a construction-general solution to the problem of the retreat from verb argument structure overgeneralization. *PLOS ONE*, 10(4), e0123723. <https://doi.org/10.1371/journal.pone.0123723>
- Arik, E. (2015). An experimental study of Turkish vowel harmony. *Poznań Studies in Contemporary Linguistics*, 51(3), 359–374. <https://doi.org/10.1515/psicl-2015-0014>
- Babel, A. M. (2025). A semiotic approach to social meaning in language. *Journal of Sociolinguistics*, 29, 59–73. <https://doi.org/10.1111/josl.12689>
- Beckner, C., Blythe, R., Bybee, J., Christiansen, M. H., Croft, W., Ellis, N. C., Holland, J., Ke, J., Larsen-Freeman, D., & Schoenemann, T. (2009). Language is a complex adaptive system: Position paper. *Language Learning*, 59(S1), 1–26. <https://doi.org/10.1111/j.1467-9922.2009.00533.x>
- Bertolotti, V. (2016). *Voseo and tuteo, the countryside and the city*. In M. I. Moyna & S. Rivera-Mills (Eds.), *Forms of address in the Spanish of the Americas* (pp. 15–34). John Benjamins Publishing Company. <https://doi.org/10.1075/ihll.10.02ber>
- Blythe, R. A., & Croft, W. (2012). S-curves and the mechanisms of propagation in language change. *Language*, 88(2), 269–304. <https://doi.org/10.1353/lan.2012.0027>
- Blythe, R. A., & Croft, W. A. (2009). The speech community in evolutionary language dynamics. *Language Learning*, 59(Suppl. 1), 47–63. <https://doi.org/10.1111/j.1467-9922.2009.00535.x>
- Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–185). MIT Press. <https://doi.org/10.7551/mitpress/6396.003.0012>
- Bybee, J. (2006). From usage to grammar: The mind's response to repetition. *Language*, 82(4), 711–733. <https://doi.org/10.1353/lan.2006.0186>
- Bybee, J. (2010). *Language, usage and cognition*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511750526>
- Bybee, J. L. (2007). *Frequency of use and the organization of language*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780195301571.001.0001>
- Cavell, S. (1969). Knowing and acknowledging. In *Must we mean what we say?* (pp. 238–266). Charles Scribner's Sons.
- Chierchia, G. (2013). *Logic in grammar: Polarity, free choice, and intervention* (Vol. 2). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199697977.001.0001>
- Chomsky, N. (1957). *Syntactic structures*. Mouton. <https://doi.org/10.1515/9783112316009>
- Chomsky, N. (1965). *Aspects of the theory of syntax*. MIT Press.
- Corbett, P. B. (2016). *Copy edit this! Quiz no. 1*. The New York Times. Retrieved December 5, 2024, from <https://www.nytimes.com/interactive/2016/11/11/insider/copy-edit-this-quiz.html>
- Croft, W. (2000). *Explaining language change: An evolutionary approach*. Longman.
- Culbertson, J., & Kirby, S. (2016). Simplicity and specificity in language: Domain-general biases have domain-specific effects. *Frontiers in Psychology*, 6, 1964. <https://doi.org/10.3389/fpsyg.2015.01964>

- Culicover, P. W., Varaschin, G., & Winkler, S. (2022). The radical unacceptability hypothesis: Accounting for unacceptability without universal constraints. *Languages*, 7(2), 96. <https://doi.org/10.3390/languages7020096>
- Cuneo, N., & Goldberg, A. E. (2023). The discourse functions of grammatical constructions explain an enduring syntactic puzzle. *Cognition*, 240, 105563. <https://doi.org/10.1016/j.cognition.2023.105563>
- Del Pinal, G. (2019). The logicity of language: A new take on triviality, “ungrammaticality”, and logical form. *Nous*, 53(4), 785–818. <https://doi.org/10.1111/nous.12235>
- Dennett, D. C. (1991). Real patterns. *The Journal of Philosophy*, 88(1), 27–51. <https://doi.org/10.2307/2027085>
- Eckert, P. (2012). Three waves of variation study: The emergence of meaning in the study of sociolinguistic variation. *Annual Review of Anthropology*, 41, 87–100. <https://doi.org/10.1146/annurev-anthro-092611-145828>
- Fanselow, G. (2021). Acceptability, grammar, and processing. In G. Goodall (Ed.), *The Cambridge handbook of experimental syntax* (pp. 118–153). Cambridge University Press. <https://doi.org/10.1017/9781108569620.006>
- Fedorenko, E., Behr, M. K., & Kanwisher, N. (2011). Functional specificity for high-level linguistic processing in the human brain. *Proceedings of the National Academy of Sciences of the United States of America*, 108(39), 16428–16433. <https://doi.org/10.1073/pnas.1112937108>
- Fedorenko, E., Duncan, J., & Kanwisher, N. (2012). Language-selective and domain-general regions lie side by side within Broca’s area. *Current Biology*, 22(21), 2059–2062. <https://doi.org/10.1016/j.cub.2012.09.011>
- Fedorenko, E., Piantadosi, S. T., & Gibson, E. A. F. (2024). Language is primarily a tool for communication rather than thought. *Nature*, 630(8017), 575–586. <https://doi.org/10.1038/s41586-024-07522-w>
- Fillmore, C. J. (1988). The mechanisms of “Construction Grammar”. *Proceedings of the Berkeley Linguistics Society*, 14, 35–55. <https://doi.org/10.3765/bls.v14i0.1794>
- Fillmore, C. J., Kay, P., & O’Connor, M. C. (1988). Regularity and idiomaticity in grammatical constructions: The case of *let alone*. *Language*, 64(3), 501–538. <https://doi.org/10.2307/414531>
- Fox, D. (2000). *Economy and semantic interpretation*. The MIT Press.
- Francis, E. J. (2022). *Gradient acceptability and linguistic theory*. Oxford University Press. <https://doi.org/10.1093/oso/9780192898944.001.0001>
- Gajewski, J. (2009). *L-triviality and grammar* [Manuscript, University of Connecticut].
- Gibson, E. (2000). The dependency locality theory: A distance-based theory of linguistic complexity. In A. Marantz, Y. Miyashita & W. O’Neil (Eds.), *Image, language, brain* (pp. 95–126). MIT Press. <https://doi.org/10.7551/mitpress/3654.003.0008>
- Gibson, E. (2026). Dependency syntax as the simplest theory of grammar. *Trends in Cognitive Sciences*. <https://doi.org/10.1016/j.tics.2026.03.002>
- Goldberg, A. E. (1995). *Constructions: A Construction Grammar approach to argument structure*. University of Chicago Press.
- Goldberg, A. E. (2011). Corpus evidence of the viability of statistical preemption. *Cognitive Linguistics*, 22(1), 131–153. <https://doi.org/10.1515/cogl.2011.006>

- Goldberg, A. E. (2019). *Explain me this: Creativity, competition, and the partial productivity of constructions*. Princeton University Press. <https://doi.org/10.1515/9780691183954>
- Goodman, N. (1955). *Fact, fiction, and forecast*. Harvard University Press.
- Grodner, D. J., & Gibson, E. (2005). Consequences of the serial nature of linguistic input for sentential complexity. *Cognitive Science*, 29(2), 261–290. https://doi.org/10.1207/s15516709cog0000_7
- Hankamer, J., & Postal, P. (1973). Whose gorilla? *Linguistic Inquiry*, 4(2), 261.
- Hart, J. T. (1965). Memory and the feeling-of-knowing experience. *Journal of Educational Psychology*, 56(4), 208–216. <https://doi.org/10.1037/h0022263>
- Haspelmath, M. (2010). Comparative concepts and descriptive categories in crosslinguistic studies. *Language*, 86(3), 663–687. <https://doi.org/10.1353/lan.2010.0021>
- Hopper, P. J. (1987). Emergent grammar. *Proceedings of the Thirteenth Annual Meeting of the Berkeley Linguistics Society*, 139–157. <https://doi.org/10.3765/bls.v13i0.1834>
- Huddleston, R., & Pullum, G. K. (2002). *The Cambridge grammar of the English language*. Cambridge University Press. <https://doi.org/10.1017/9781316423530>
- Kabak, B., & Vogel, I. (2001). The phonological word and stress assignment in Turkish. *Phonology*, 18(3), 315–360. <https://doi.org/10.1017/S0952675701004201>
- Kay, P., & Fillmore, C. J. (1999). Grammatical constructions and linguistic generalizations: The *What's x doing y?* construction. *Language*, 75(1), 1–33. <https://doi.org/10.2307/417472>
- Keller, R. (1994). *On language change: The invisible hand in language* [Translated by Brigitte Nerlich]. Routledge.
- Kilani-Schoch, M., & Dressler, W. U. (2005). *Morphologie naturelle et flexion du verbe français*. Narr.
- Kilgariff, A. (2004). How dominant is the commonest sense of a word? In P. Sojka, I. Kopeček & K. Pala (Eds.), *Text, speech, dialogue*. https://doi.org/10.1007/978-3-540-30120-2_14
- Kirby, S., Griffiths, T., & Smith, K. (2014). Iterated learning and the evolution of language. *Current Opinion in Neurobiology*, 28, 108–114. <https://doi.org/10.1016/j.conb.2014.07.014>
- Kroch, A. S. (1989). Reflexes of grammar in patterns of language change. *Language Variation and Change*, 1(3), 199–244. <https://doi.org/10.1017/S0954394500000168>
- Labov, W. (1972). *Sociolinguistic patterns*. University of Pennsylvania Press.
- Lakoff, G. (1971). On generative semantics. In D. D. Steinberg & L. A. Jakobovits (Eds.), *Semantics: An interdisciplinary reader in philosophy, linguistics, and psychology* (pp. 232–296). Cambridge University Press.
- Lazzaro, J. C. (2026). *Nonlinear regression discontinuity designs with covariates* [Job-market paper, University of Wisconsin–Madison; version of 15 April 2026]. Retrieved July 1, 2026, from <https://econ.wisc.edu/staff/lazzaro-john/>
- Lu, J., Frank, M., & Degen, J. (2024). A meta-analysis of syntactic satiation in extraction from islands. *Glossa Psycholinguistics*, 3(1), 1–33. <https://doi.org/10.5070/g60111425>
- McCawley, J. D. (1968). The role of semantics in a grammar. In E. Bach & R. T. Harms (Eds.), *Universals in linguistic theory* (pp. 125–170). Holt, Rinehart; Winston.
- McCawley, J. D. (1974). Interview with James McCawley. In H. Parret (Ed.), *Discussing language: Dialogues with Wallace L. Chafe, Noam Chomsky, Algirdas J. Greimas, M. A. K. Halliday, Peter Hartmann, George Lakoff, Sydney M. Lamb, André Martinet, James McCawley, Se-*

- bastian K. Saumjan, Jacques Bouveresse* (Vol. 93). De Gruyter Mouton. <https://doi.org/10.1515/9783110813456-010>
- Morgan, J. L. (1973). Sentence fragments and the notion ‘sentence’. In T. M. Lightner & J. D. Macaulay (Eds.), *Issues in linguistics: Papers in honor of Henry and Renée Kahane* (pp. 719–751). University of Illinois Press.
- Morin, C., & Grieve, J. (2024). The semantics, sociolinguistics, and origins of double modals in American English: New insights from social media. *PLoS ONE*, 19(1), e0295799. <https://doi.org/10.1371/journal.pone.0295799>
- Oomen, C. C. E., Postma, A., & Kolk, H. H. J. (2005). Speech monitoring in aphasia: Error detection and repair behaviour in a patient with Broca’s aphasia. In R. J. Hartsuiker, R. Bastiaanse, A. Postma & F. Wijnen (Eds.), *Phonological encoding and monitoring in normal and pathological speech* (pp. 203–222). Psychology Press.
- Perfors, A., Tenenbaum, J. B., & Regier, T. (2011). The learnability of abstract syntactic principles. *Cognition*, 118(3), 306–338. <https://doi.org/10.1016/j.cognition.2010.11.001>
- Post, E. L. (1943). Formal reductions of the general combinatorial decision problem. *American Journal of Mathematics*, 65(2), 197–215. <https://doi.org/10.2307/2371809>
- Prince, A., & Smolensky, P. (2004). *Optimality Theory: Constraint interaction in generative grammar*. Blackwell Publishing. <https://doi.org/10.1002/9780470759400>
- Pullum, G. K. (2009). More people than you think will understand. Retrieved December 5, 2024, from <https://languagelog.ldc.upenn.edu/nll/?p=2013>
- Reynolds, B. (2024a). *Not every Whose down in Who-ville likes appearing a lot: Pragmatic constraints on independent relative Whose* [Manuscript, LingBuzz/008313]. <https://ling.auf.net/lingbuzz/008313>
- Reynolds, B. (2024b). Why *more* and *less* are never adverbs. *Journal of Linguistics*, 1–29. <https://doi.org/10.1017/S0022226724000239>
- Reynolds, B. (2026a). *Kinds as projectibility profiles: Support grades and demotion rules* [Manuscript archived at PhilArchive, 14 June 2026]. <https://philarchive.org/rec/REYKAP-2>
- Reynolds, B. (2026b). *Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds* [Manuscript archived at PhilArchive, 1 June 2026]. <https://philarchive.org/rec/REYNES>
- Reynolds, B. (2026c). *Why clause structure is judged like tense and agreement: Public-update operators and grammaticality* [Preprint, LingBuzz/009706]. <https://ling.auf.net/lingbuzz/009706>
- Reynolds, B. (2026d). *Why English doesn’t extract left branches (yet)* [Preprint, LingBuzz/009708]. <https://ling.auf.net/lingbuzz/009708>
- Ritchie, G. D., & Thompson, H. S. (1984). Natural language processing. In T. O’Shea & M. Eisenstadt (Eds.), *Artificial intelligence: Tools, techniques and applications* (pp. 358–388). Harper; Row.
- Sag, I. A. (2012). Sign-Based Construction Grammar: An informal synopsis. In H. C. Boas & I. A. Sag (Eds.), *Sign-Based Construction Grammar* (pp. 69–202). CSLI Publications.
- Sapir, E. (1921). *Language: An introduction to the study of speech*. Harcourt Brace & Company.
- Sapir, E. (1927). The unconscious patterning of behavior in society [Originally published 1927 in Ethel S. Dummer, ed., *The Unconscious: A Symposium*. New York, Knopf: 114–142. Reprinted 1951]. In D. G. Mandelbaum (Ed.), *Selected writings of Edward Sapir in language*,

- culture and personality* (pp. 544–559). University of California Press. <https://doi.org/10.1037/13401-006>
- Saussure, F. d. (1916). *Cours de linguistique générale* [Edited by Charles Bally and Albert Sechehaye. English translation: *Course in General Linguistics*, trans. Wade Baskin, New York: Philosophical Library, 1959]. Payot.
- Schmid, H.-J. (2020). *The dynamics of the linguistic system: Usage, conventionalization, and entrenchment*. Oxford University Press. <https://doi.org/10.1093/oso/9780198814771.001.0001>
- Schütze, C. T. (2016). *The empirical base of linguistics: Grammaticality judgments and linguistic methodology*. Language Science Press. <https://doi.org/10.17169/langsci.b89.100>
- Scott-Phillips, T. (2026). Communication and grammar: A synthesis. *Psychological Review*, 133(1), 81–93. <https://doi.org/10.1037/rev0000542>
- Selinker, L. (1972). Interlanguage. *International Review of Applied Linguistics in Language Teaching*, 10(1-4), 209–232. <https://doi.org/10.1515/iral.1972.10.1-4.209>
- Sezer, E. (1981). Vowel harmony in Turkish: Evidence for feature weighting. In D. I. Slobin & K. Zimmer (Eds.), *Linguistic studies in Turkish and Turkic* (pp. 213–235). John Benjamins.
- Shain, C., Blank, I. A., van Schijndel, M., Schuler, W., & Fedorenko, E. (2020). fMRI reveals language-specific predictive coding during naturalistic sentence comprehension. *Neuropsychologia*, 138. <https://doi.org/10.1016/j.neuropsychologia.2019.107307>
- Silverstein, M. (1976). Shifters, linguistic categories, and cultural description. In K. H. Basso & H. A. Selby (Eds.), *Meaning in anthropology* (pp. 11–55). University of New Mexico Press.
- Snyder, W. (2022). On the nature of syntactic satiation. *Languages*, 7(1), 38. <https://doi.org/10.3390/languages7010038>
- Stefanowitsch, A. (2006). Negative evidence and the raw frequency fallacy. *Corpus Linguistics and Linguistic Theory*, 2(1), 61–77. <https://doi.org/10.1515/CLLT.2006.003>
- Techio, J. (2026, March). *The claim upon the training data* [Essay. Engages Cavell’s knowledge/acknowledgment distinction with LLMs and Cowen’s GOAT]. <https://www.jonadas.com/writing/essays/the-claim-upon-the-training-data>
- Toribio, A. J. (2001). Accessing bilingual code-switching competence. *International Journal of Bilingualism*, 5(4), 403–436. <https://doi.org/10.1177/13670069010050040201>
- Verkerk, A., Shcherbakova, O., Haynie, H. J., Skirgård, H., Rzymiski, C., Atkinson, Q. D., Greenhill, S. J., & Gray, R. D. (2025). Enduring constraints on grammar revealed by Bayesian spatio-phylogenetic analyses. *Nature Human Behaviour*. <https://doi.org/10.1038/s41562-025-02325-z>
- Weissweiler, L., He, T., Otani, N., Mortensen, D. R., Levin, L., & Schütze, H. (2023). Construction grammar provides unique insight into neural language models. *arXiv preprint arXiv:2302.02178*.
- Wiese, H. (2023). *Grammatical systems without language borders: Lessons from free-range language*. Language Science Press. <https://doi.org/10.5281/zenodo.10276182>